TOPIC 2

Gradient Descent Maths and Calculus: Explained

The original class slides, with a plain explanation under each one

> **NOTE** **How to read this document**
>
> - Every original slide is kept exactly as it was, with its slide number under it
> - Under each slide (or small set of slides) you get a plain explanation: **what you are looking at** and **what it means**
> - The slides come from a recorded class, so a few look like screenshots of a video: that is normal

These 171 slides walk through the **maths of gradient descent** with a tiny example: three people's weights and heights, and one unknown number, the intercept of a line. Nothing is skipped: every step of the calculation is shown on its own slide.

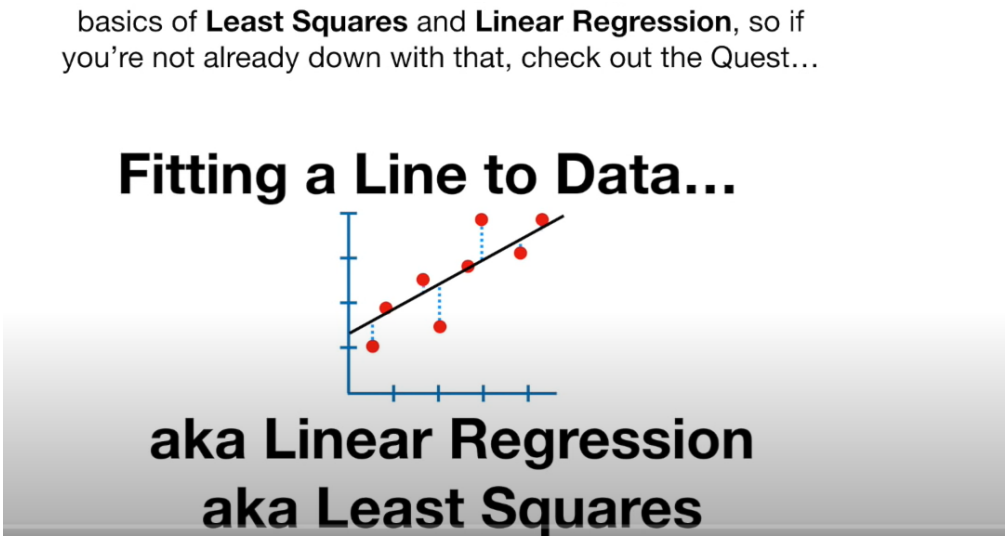
The slides come from a step-by-step animated lesson, so many slides look almost identical: they are consecutive frames where one new thing appears at a time. In this document such frames are grouped together and explained as one idea.

The story in five lines: (1) guess an intercept; (2) measure how wrong the line is; (3) use calculus to find the slope of the error curve; (4) step downhill by slope × learning rate; (5) repeat until the steps become tiny.

Words you will meet

Word	Plain meaning
Residual	Real value minus predicted value for one data point
SSR	Sum of Squared Residuals: square every residual and add them up. Smaller is better
Loss function	A formula that scores how wrong the model is (SSR is one)
Derivative	The slope of a curve at one point. Steep means far from the minimum; near zero means at the minimum
Chain rule	A maths rule for differentiating a function inside another function
Step size	slope × learning rate: how far we move this time
Learning rate	A small number you choose (here 0.1) that scales every step
Intercept	Where the line crosses the vertical axis
Least Squares	A different method that solves for the best line directly; used here to check the answer

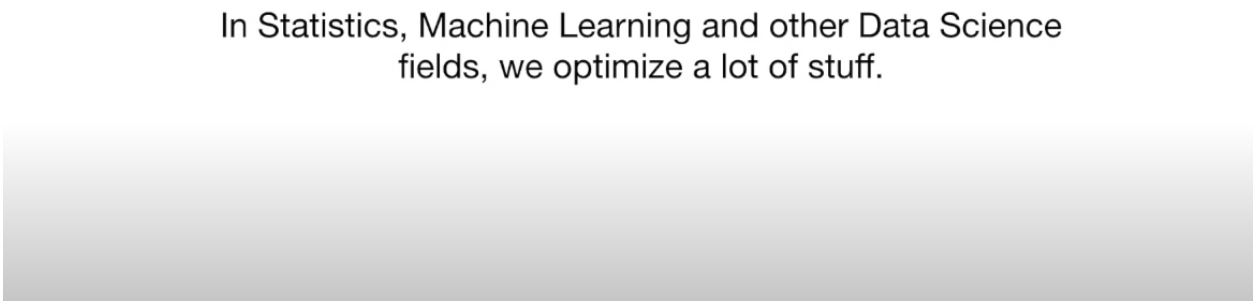
Slide 1: Where this video starts: fitting a line to data



Slide 1

The title slide says **Fitting a Line to Data**, also known as **linear regression** or **least squares**. Those three names mean the same thing: drawing the straight line that best matches a cloud of points. The small note on top says that if you do not know these basics yet, you should learn them first.

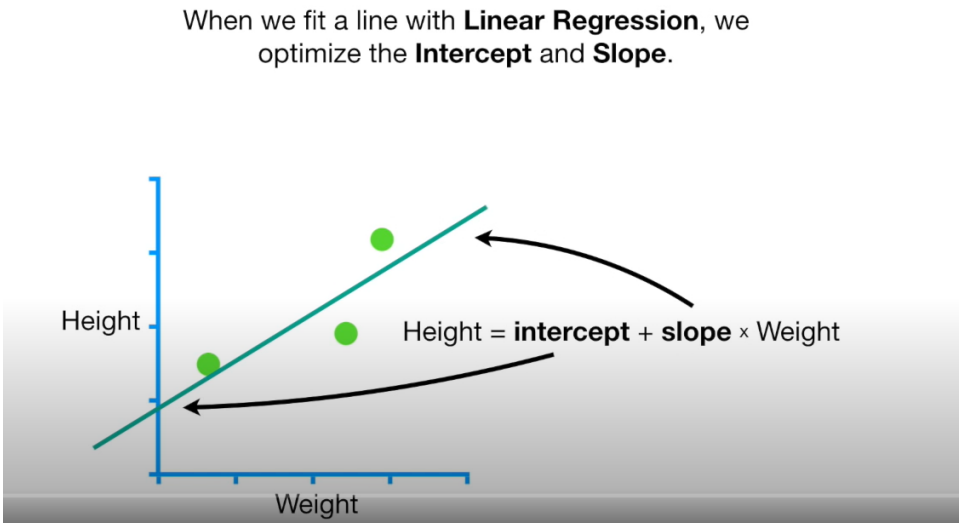
Slide 2: The big claim: statistics and ML optimise a lot of things



Slide 2

Optimise means 'find the best version of something'. In statistics and machine learning we constantly search for the best values of some numbers. The next few slides show examples of what gets optimised.

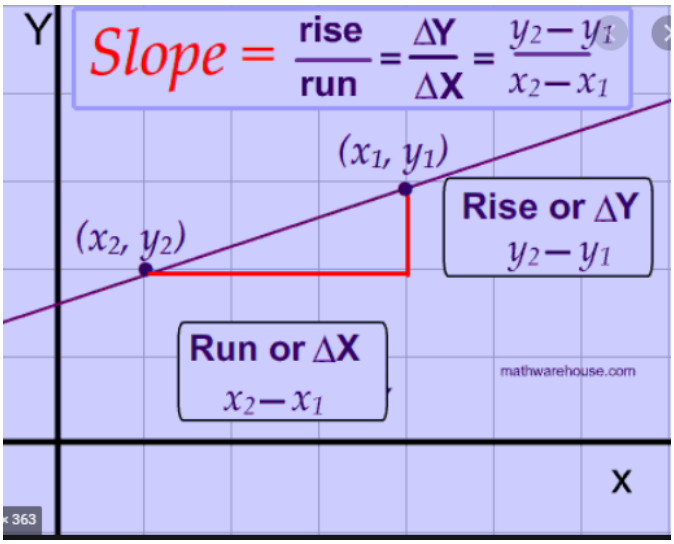
Slide 3: Example 1: optimising a line (linear regression)



Slide 3

When we fit a line to data we optimise two numbers: the **intercept** and the **slope**. The line is written **Height = intercept + slope × Weight**: for a given weight it predicts a height. The two arrows point to where each number appears: the intercept is where the line starts on the vertical axis, the slope is how steeply it climbs.

Slide 4: What slope means

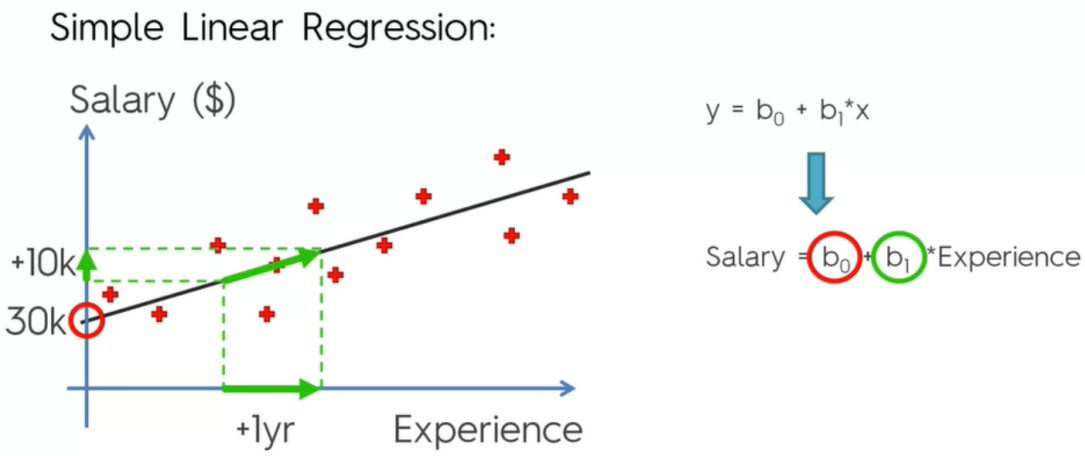


Slide 4

A reminder from school maths. Slope is **rise divided by run**: pick two points on the line, find how far up you go (rise, ΔY = y2 − y1) and how far across (run, ΔX = x2 − x1), then divide. A steep line has a big slope; a flat line has a slope near 0.

$$\text{slope} = \frac{\text{rise}}{\text{run}} = \frac{\Delta Y}{\Delta X} = \frac{y_2 - y_1}{x_2 - x_1}$$

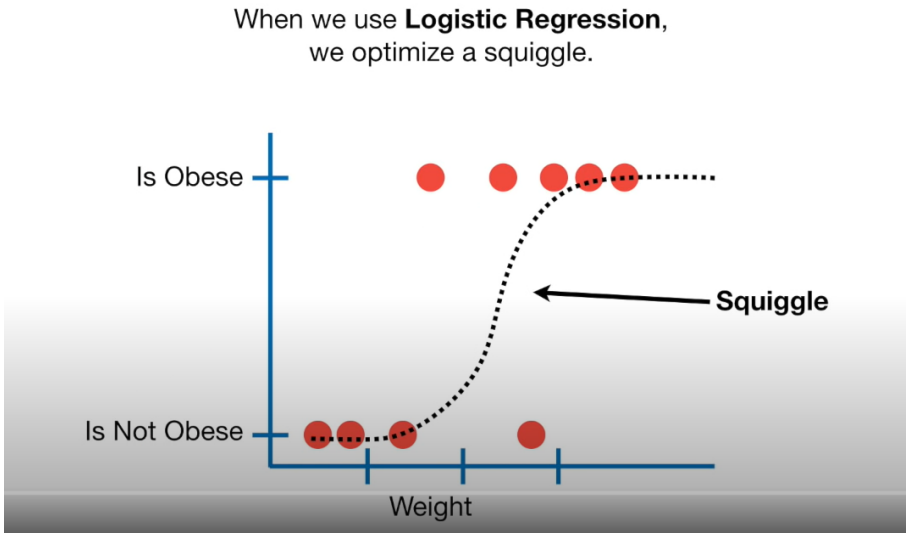
Slide 5: A real-life reading of the line



Slide 5

Simple linear regression: salary against years of experience. The formula $y = b_0 + b_1x$ becomes **Salary = b_0 + b_1 × Experience**. The intercept b_0 (circled red, about 30k) is the salary with no experience. The slope b_1 (circled green) is the extra salary for each extra year: here each +1 year adds about +10k.

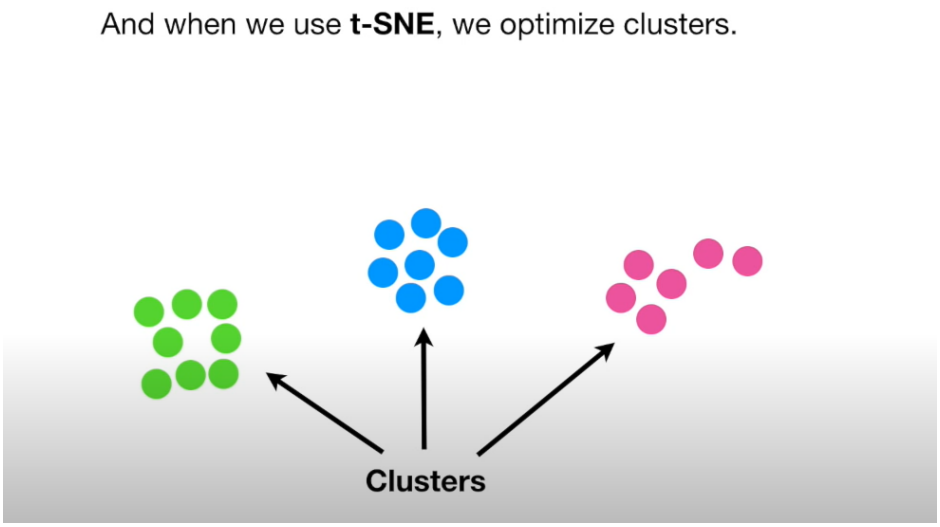
Slide 6: Example 2: optimising a squiggle (logistic regression)



Slide 6

In **logistic regression** the model is not a straight line but an S-shaped curve (the squiggle). Here it predicts whether a person **is obese or not** from their weight: dots at the top are 'obese', dots at the bottom are 'not obese'. Fitting the squiggle to the dots is also an optimisation problem.

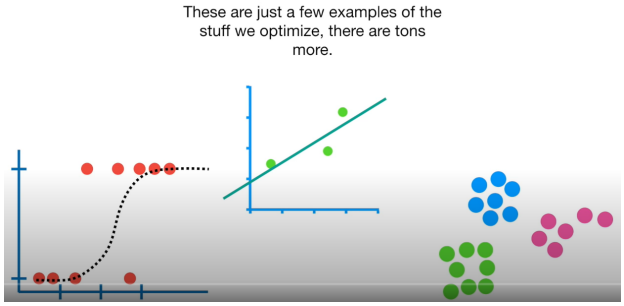
Slide 7: Example 3: optimising clusters (t-SNE)



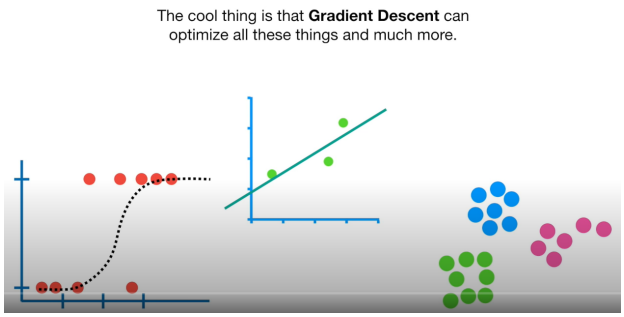
Slide 7

t-SNE is a method that arranges points into groups called **clusters** (the green, blue and pink groups). Finding the best arrangement is yet another thing that is optimised.

Slides 8 to 9: Gradient descent can optimise all of these



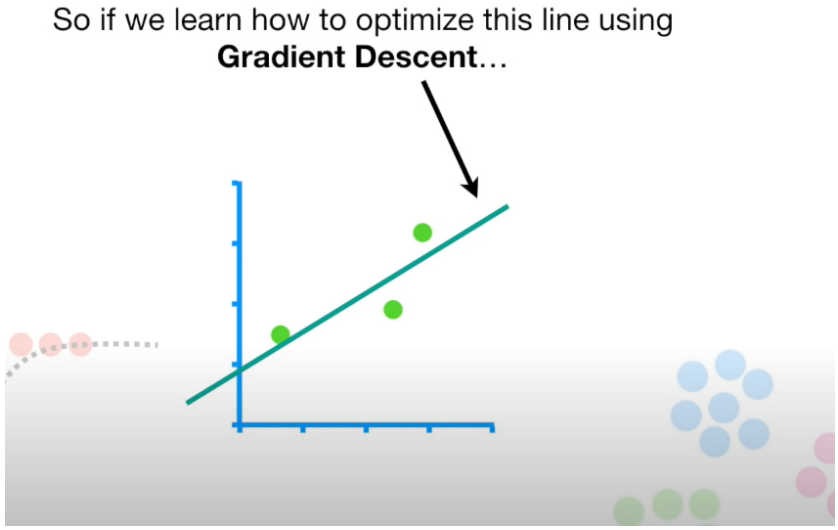
Slide 8



Slide 9

These are only a few examples; there are many more. The 'cool thing' is that one method, **gradient descent**, can optimise all of them. So learning it once is very valuable.

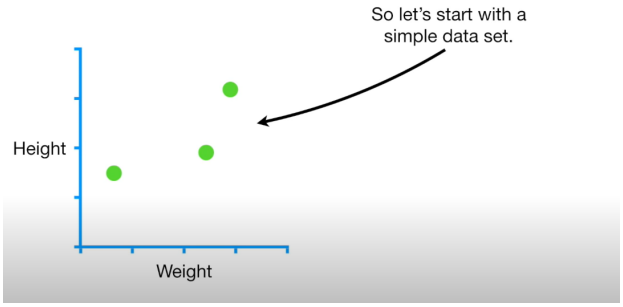
Slide 10: Plan: learn gradient descent on the simplest case



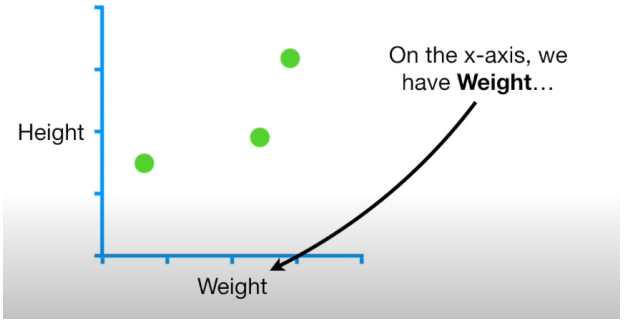
Slide 10

To learn gradient descent we start with the easiest example: optimising a straight line.

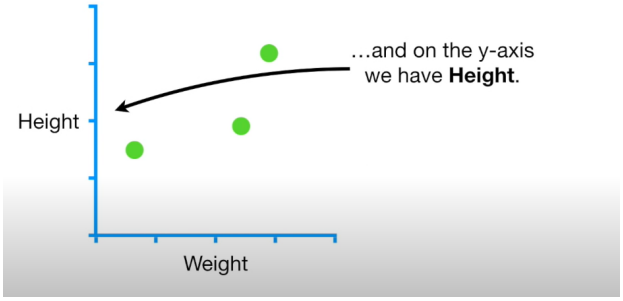
Slides 11 to 13: The tiny data set



Slide 11



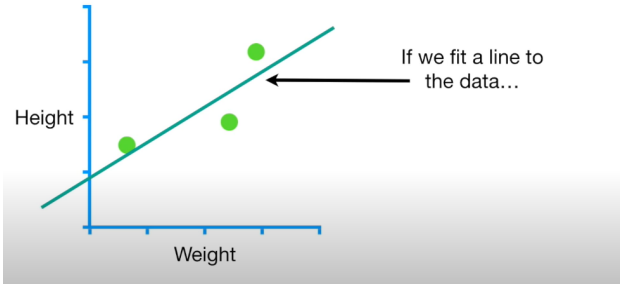
Slide 12



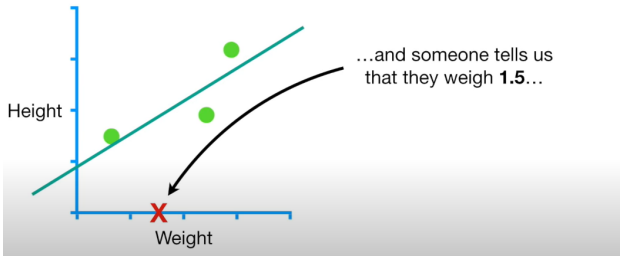
Slide 13

A very small data set: three green dots. The horizontal axis is **Weight** and the vertical axis is **Height** of three people. Three points are enough to see every idea clearly.

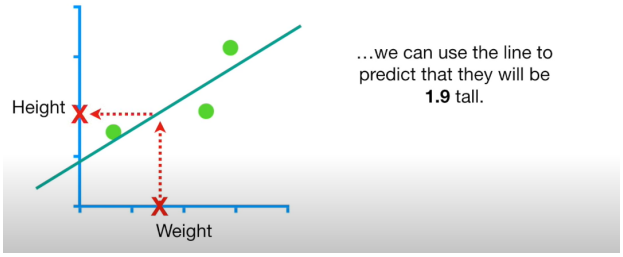
Slides 14 to 17: What a fitted line is good for: prediction



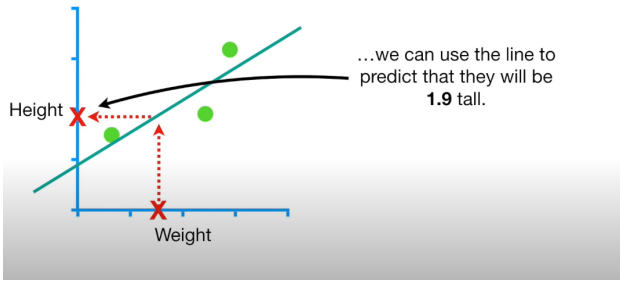
Slide 14



Slide 15



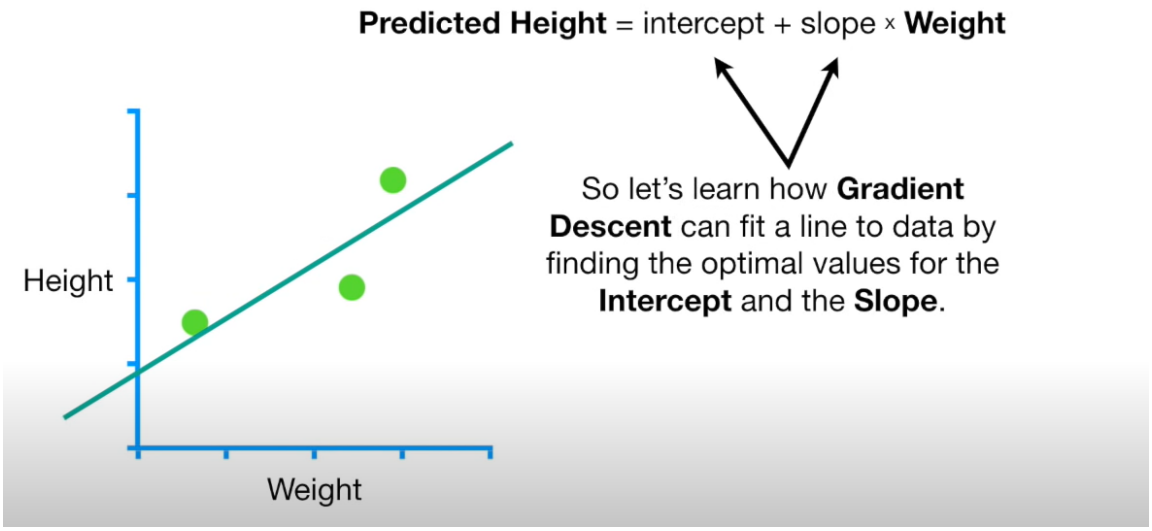
Slide 16



Slide 17

Once a line is fitted to the data (slide 14), we can use it to make predictions. If someone tells us they weigh **1.5**, we find 1.5 on the weight axis, go up to the line, and read across: the line predicts a height of **1.9**.

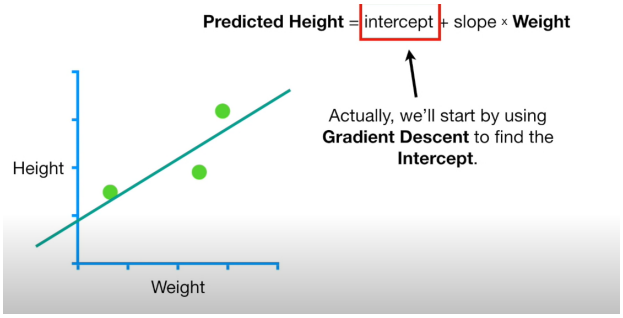
Slide 18: The goal of this video



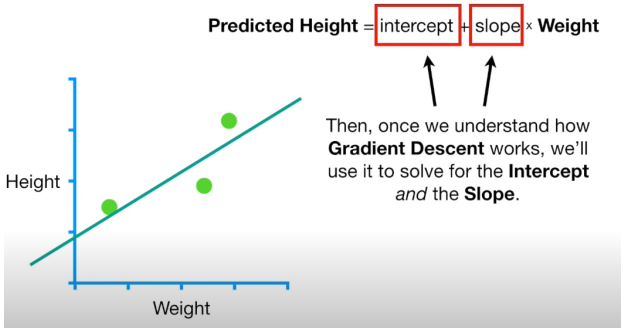
Slide 18

Learn how gradient descent can fit a line to data by finding the best values for the **intercept** and the **slope**. The formula in the title, **Predicted Height = intercept + slope × Weight**, is the line we want to fit.

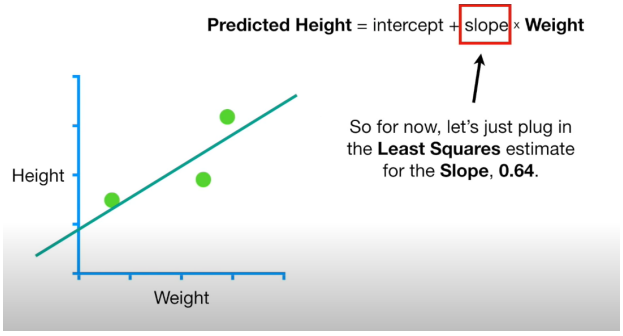
Slides 19 to 21: Divide the problem: intercept first, slope later



Slide 19



Slide 20

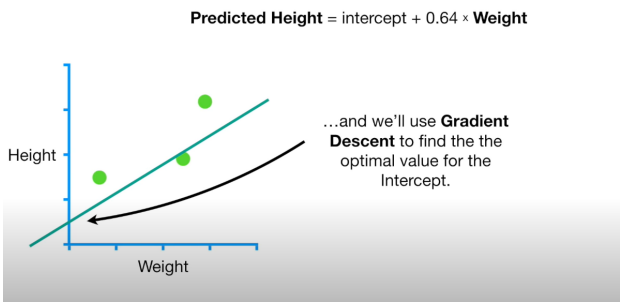


Slide 21

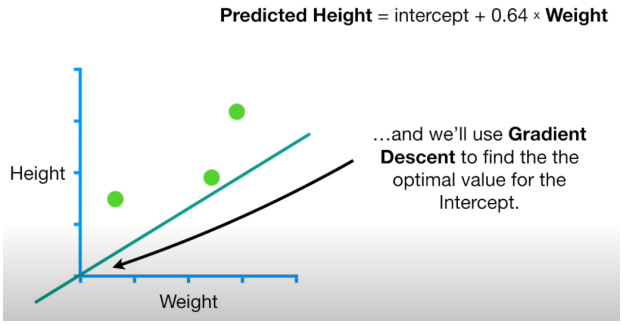
To keep things simple, the video optimises only **one** number at a time. First: use gradient descent to find the **intercept**. Later, once the method is understood, the same trick is used for the intercept and the slope together.

- The slope is fixed for now. A different method (Least Squares) already told us a good slope: **0.64**
- Slide 21: 'let's just plug in the Least Squares estimate for the slope'

Slides 22 to 23: Slope fixed at 0.64: only the intercept is unknown



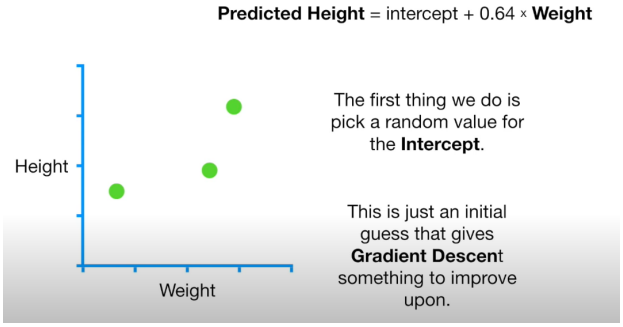
Slide 22



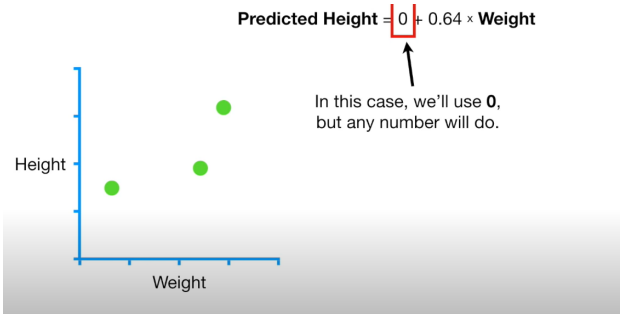
Slide 23

Now the formula reads **Predicted Height = intercept + 0.64 × Weight**. Only the intercept is unknown, so gradient descent has just one number to find. The arrow points to where the intercept shows up: it decides how high or low the line sits.

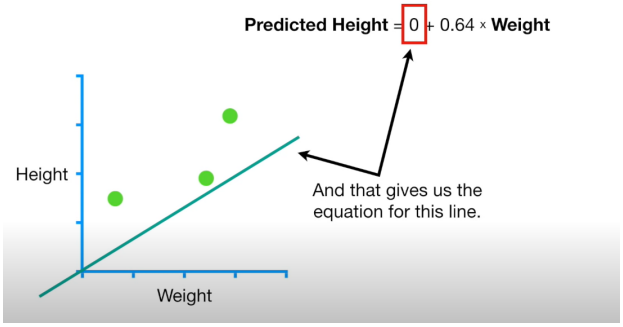
Slides 24 to 26: Step 1: start with a random guess



Slide 24



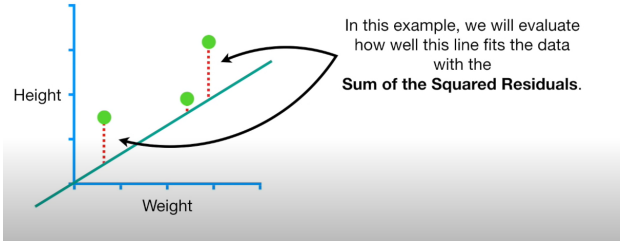
Slide 25



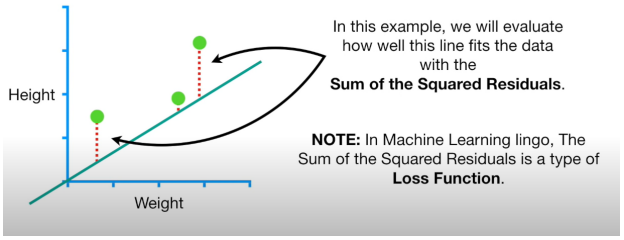
Slide 26

Gradient descent always begins with a guess. The first guess for the intercept is chosen randomly. Here the video uses **0** (but any number would do). It is only a starting point for the method to improve. With intercept 0 the line starts at the origin, giving the equation **Predicted Height = 0 + 0.64 × Weight**.

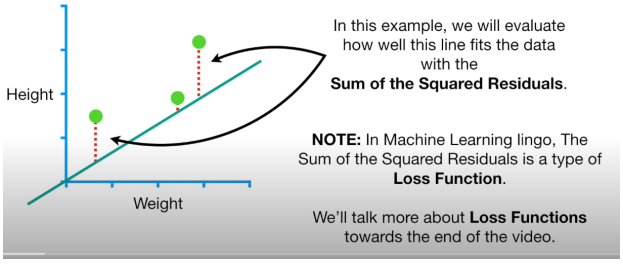
Slides 27 to 29: How do we judge a guess? Sum of Squared Residuals



Slide 27



Slide 28

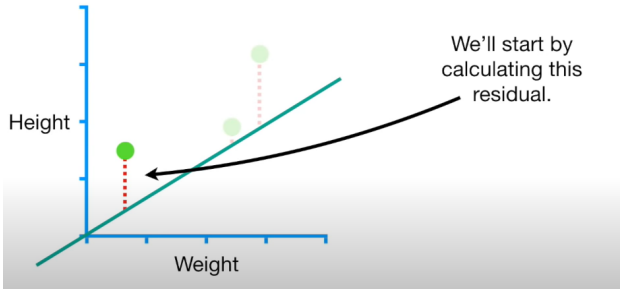


Slide 29

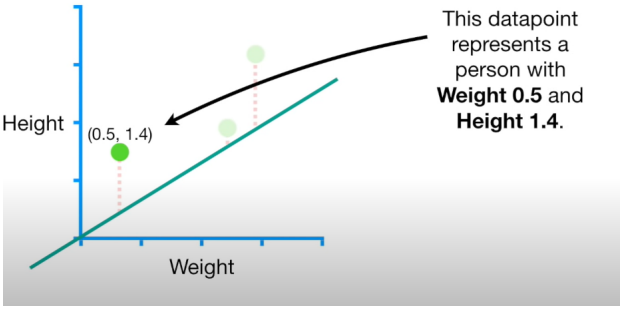
To know whether a line is good, we measure how far the data points are from it. The dotted red lines are those distances. Adding up the squares of those distances gives a single score called the **Sum of the Squared Residuals** (SSR): lower is better.

- In machine learning this kind of score is called a **loss function** (also cost function)
- Loss functions are covered at the end of the video

Slides 30 to 31: Start with one point: the residual



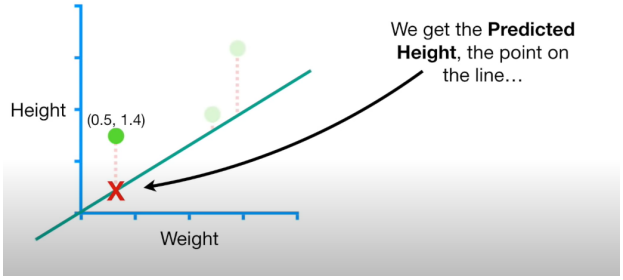
Slide 30



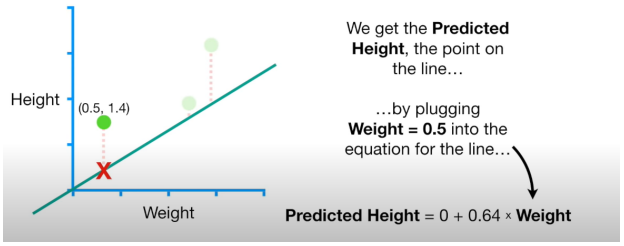
Slide 31

We calculate the distance for one point at a time. The first data point is at **Weight 0.5, Height 1.4**. The residual of a point is the gap between where the point actually is and where the line says it should be.

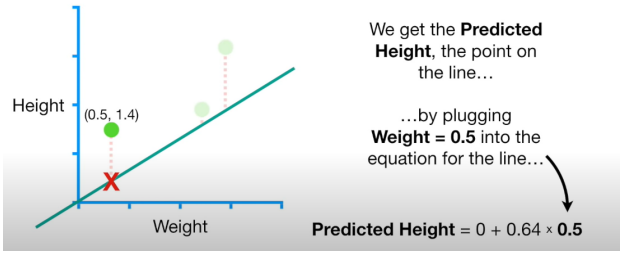
Slides 32 to 35: Step: get the line's prediction for that point



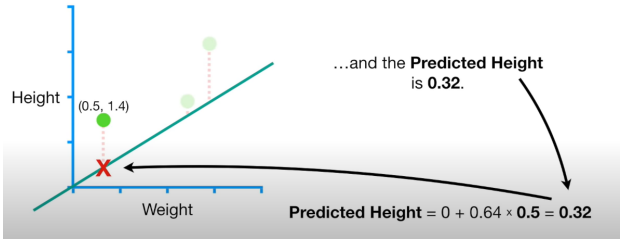
Slide 32



Slide 33



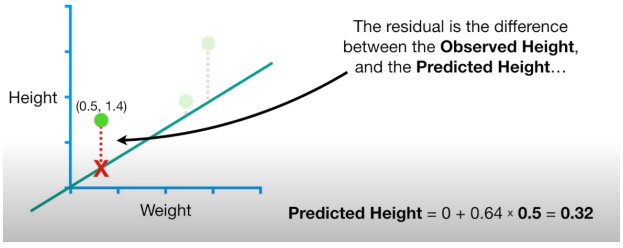
Slide 34



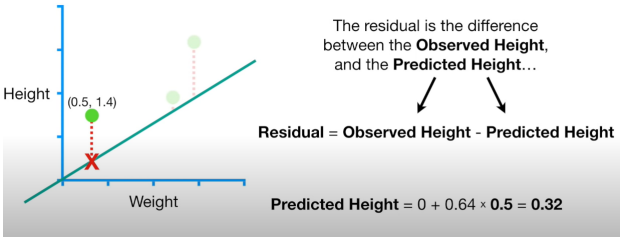
Slide 35

To find where the line is at Weight = 0.5, plug 0.5 into the line's equation: Predicted Height = 0 + 0.64 × 0.5 = **0.32**. That is the point on the line (the red X).

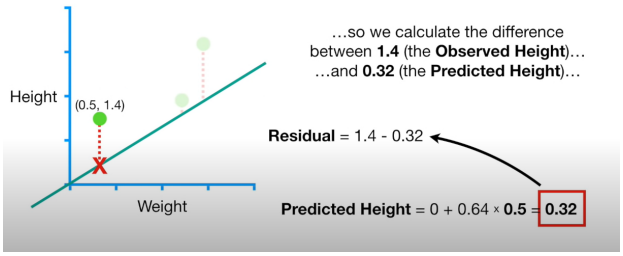
Slides 36 to 39: Step: compute the residual



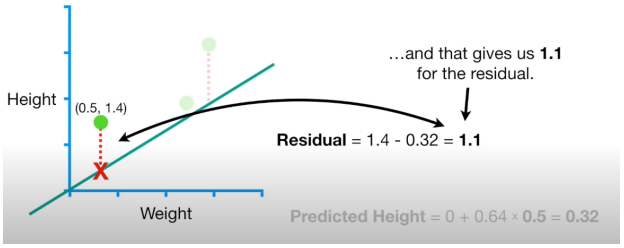
Slide 36



Slide 37



Slide 38

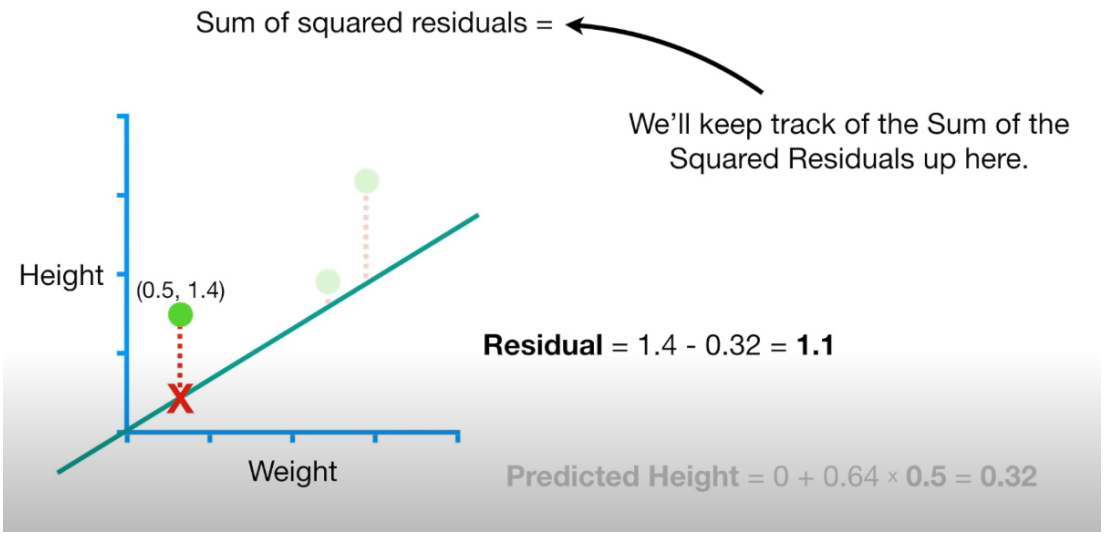


Slide 39

Residual = Observed Height − Predicted Height. The observed (real) height is 1.4 and the predicted height is 0.32, so the residual is 1.4 − 0.32 = **1.1**. The line is 1.1 too low at this point.

$$\text{Residual} = \text{Observed} - \text{Predicted} = 1.4 - 0.32 = 1.1$$

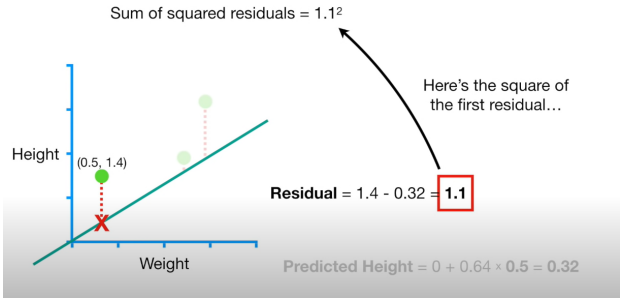
Slide 40: Keep a running total



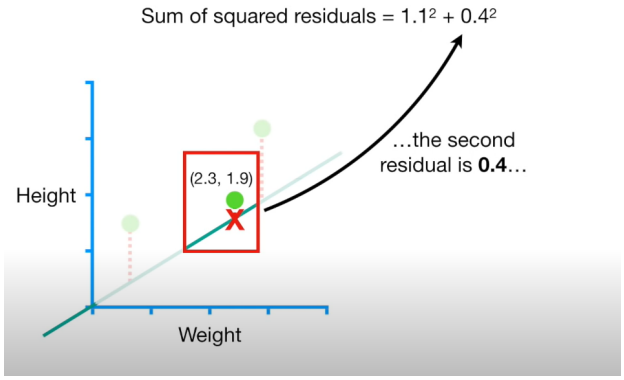
Slide 40

We start a running total called the **Sum of squared residuals**. The first point has residual 1.1, so its contribution will be 1.1^2 (the next slides do this squaring and then add the other two points).

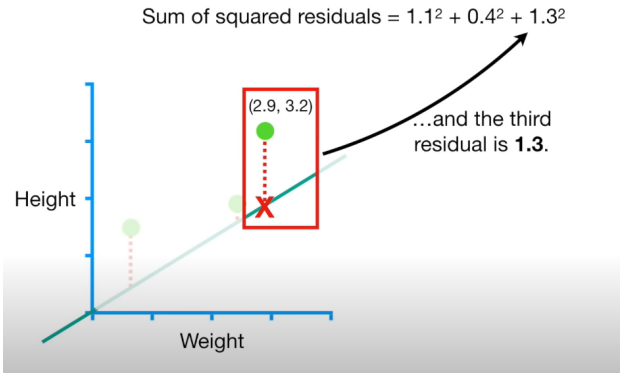
Slides 41 to 44: Adding up the squared residuals



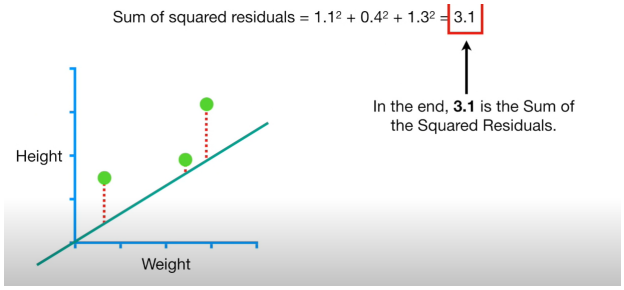
Slide 41



Slide 42



Slide 43

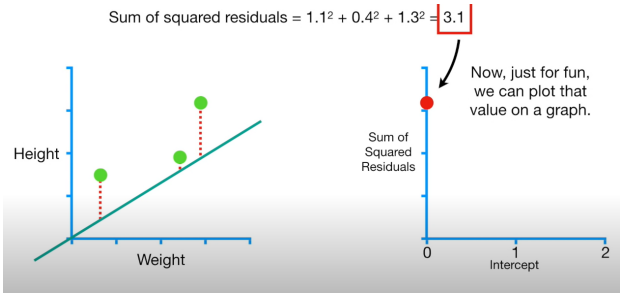


Slide 44

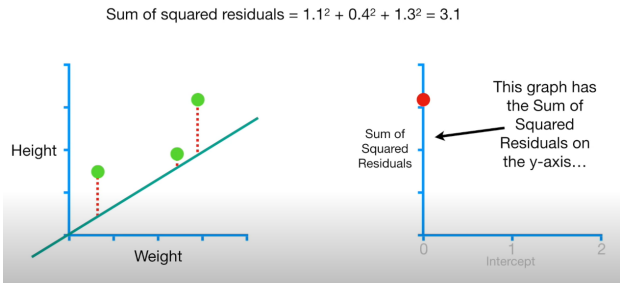
Each point's residual is squared and added to the running total. First residual 1.1, second 0.4 (point at weight 2.3, height 1.9), third 1.3 (point at weight 2.9, height 3.2). The total is **$1.1^2 + 0.4^2 + 1.3^2 = 3.1$**. So with intercept 0, the line has a Sum of Squared Residuals of 3.1.

$$SSR = 1.1^2 + 0.4^2 + 1.3^2 = 3.1$$

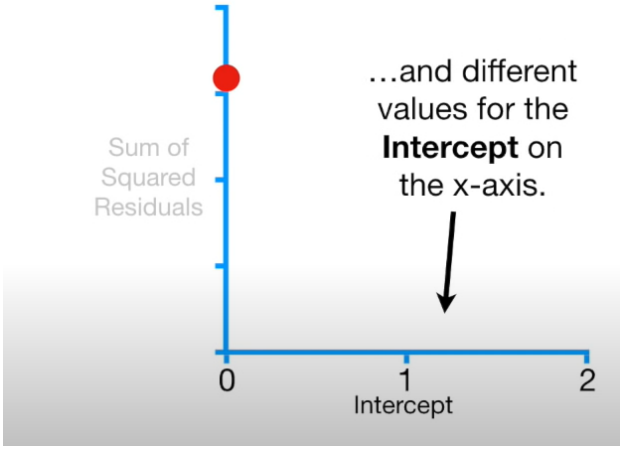
Slides 45 to 48: Plot that score on a new graph



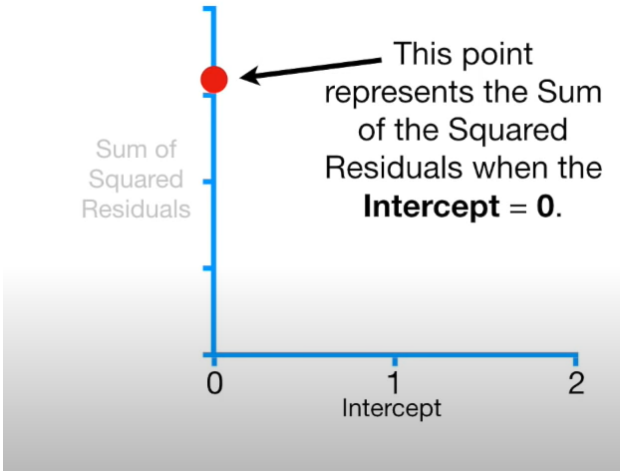
Slide 45



Slide 46



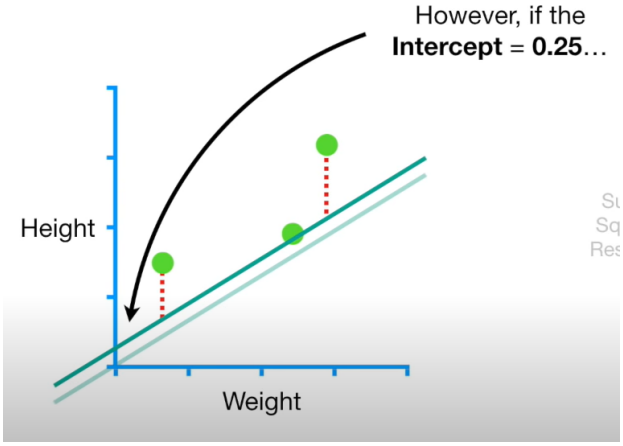
Slide 47



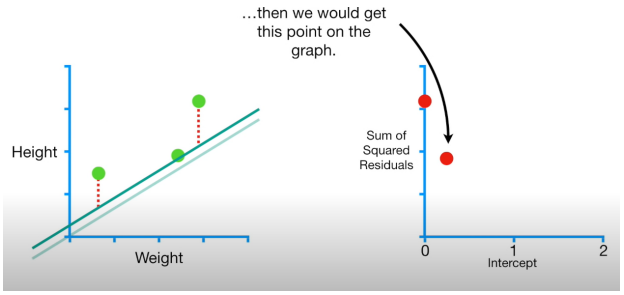
Slide 48

Just for fun, the score 3.1 is plotted as a red dot on a new graph. The vertical axis is the **Sum of Squared Residuals** and the horizontal axis is the **intercept**. This single red dot says: *when the intercept is 0, the SSR is 3.1.* Different intercepts will give different points on this graph.

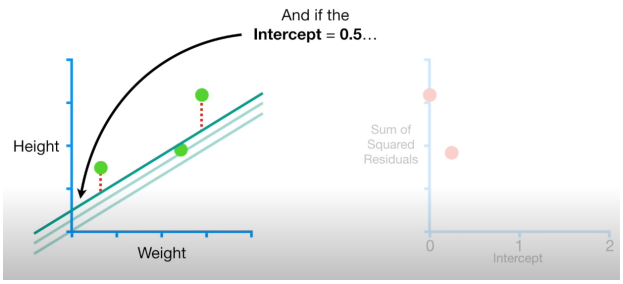
Slides 49 to 52: Try another intercept: 0.25



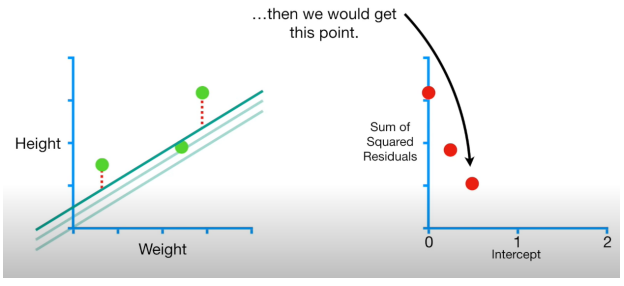
Slide 49



Slide 50



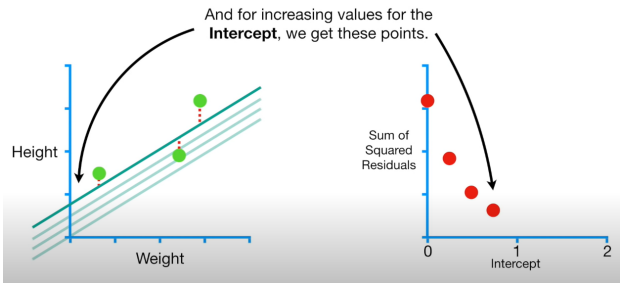
Slide 51



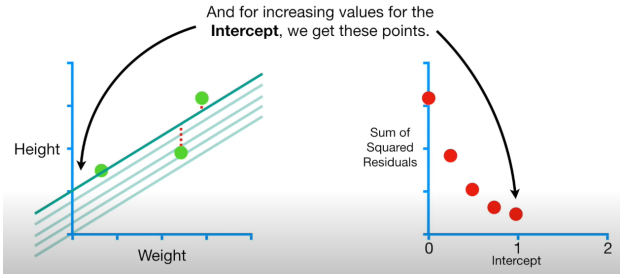
Slide 52

If the intercept is 0.25, the line shifts up a little, closer to the dots. The residuals (dotted red lines) get shorter, so the SSR goes down, and a new red dot appears lower on the graph. Then the same is done for intercept 0.5, giving another lower point.

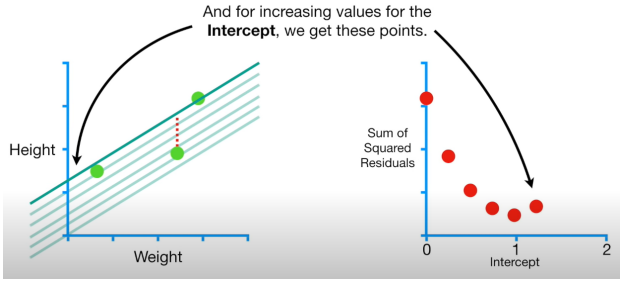
Slides 53 to 57: Many intercepts make a curve



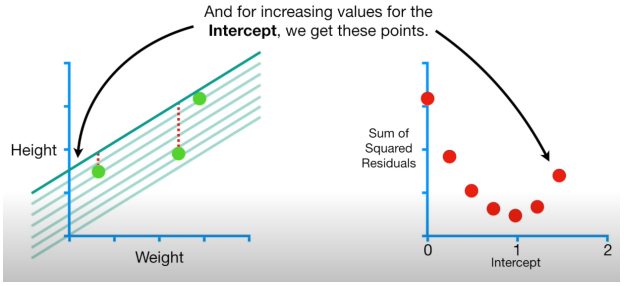
Slide 53



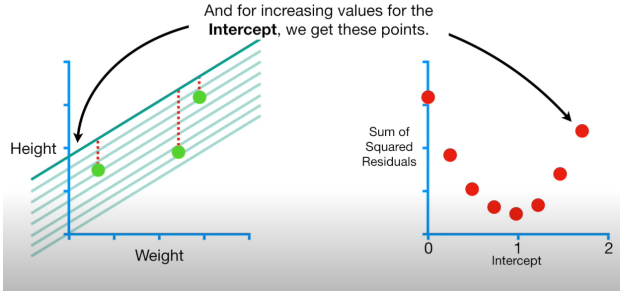
Slide 54



Slide 55



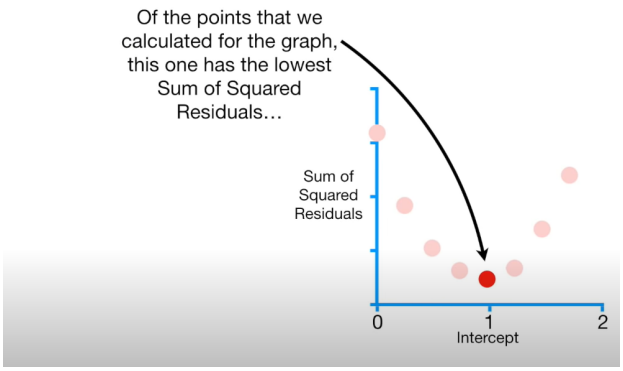
Slide 56



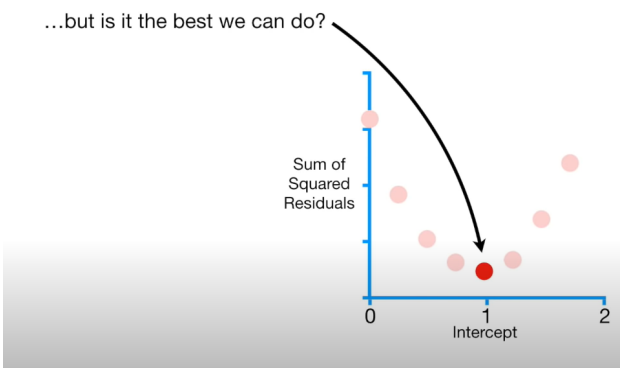
Slide 57

Repeating this for increasing intercept values gives more and more points. The dots form a **U shape (a bowl)**: the SSR falls as the line moves towards the data, reaches a lowest point, then rises again once the line goes past the data. This bowl is the same picture as the cost curve from the first topic.

Slides 58 to 59: Which point is best so far?



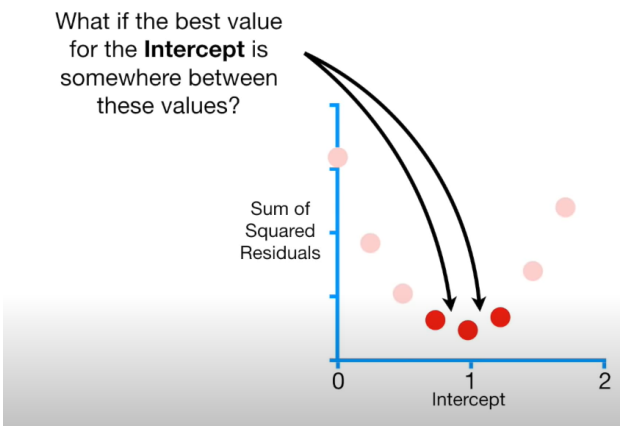
Slide 58



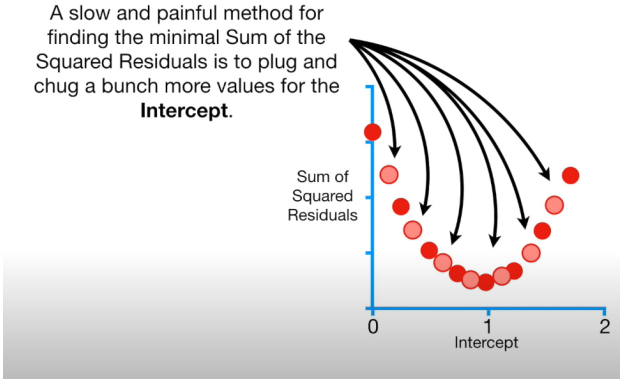
Slide 59

Out of all the points we calculated, the one near intercept 1 has the lowest Sum of Squared Residuals. But the question is: **is it the best we can do?**

Slides 60 to 61: The best value may be between the points we tried



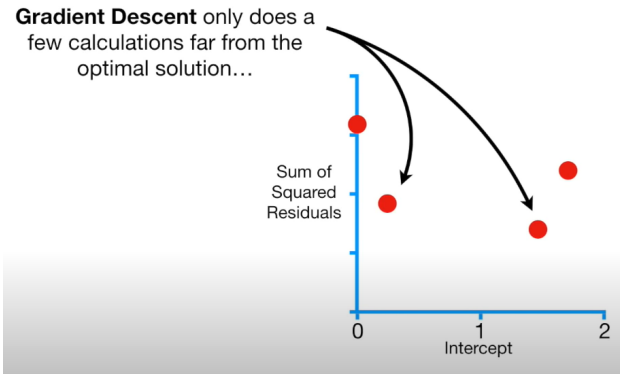
Slide 60



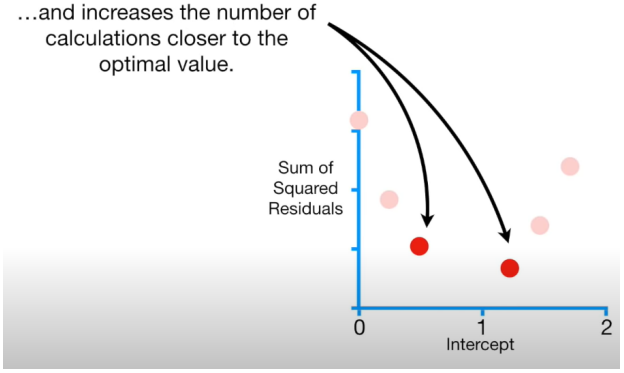
Slide 61

The true best intercept could easily sit *between* two of the values we tested. One way to find it is the slow and painful method: plug in lots more intercept values, one after another (this is called trial and error, or 'plug and chug').

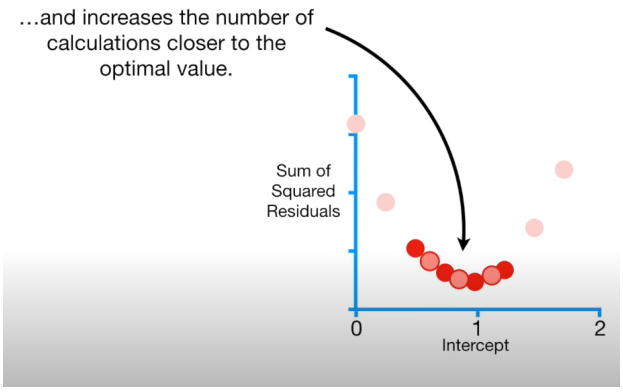
Slides 62 to 64: Gradient descent takes fewer, smarter steps



Slide 62



Slide 63

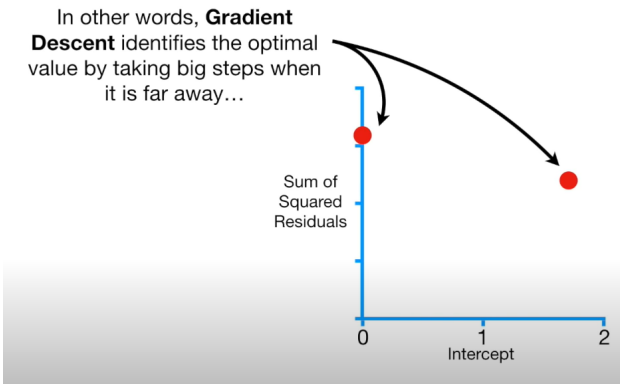


Slide 64

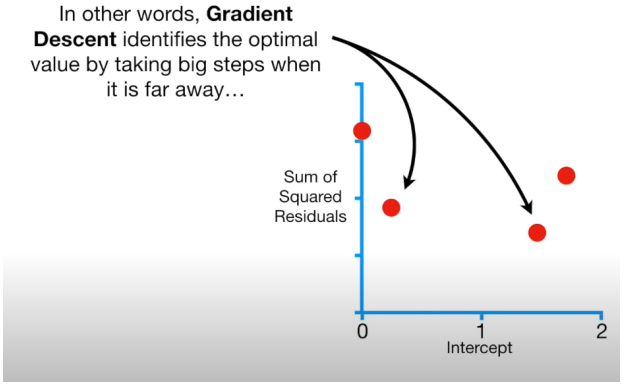
Gradient descent does only a few calculations when it is far from the best solution, and then does more calculations only when it gets close to the optimal value. So it does not waste time testing hundreds of points: it takes big steps at first, then smaller and smaller steps as it nears the bottom of the bowl.

- Fewer calculations = faster
- Spends its effort only where precision matters (near the minimum)

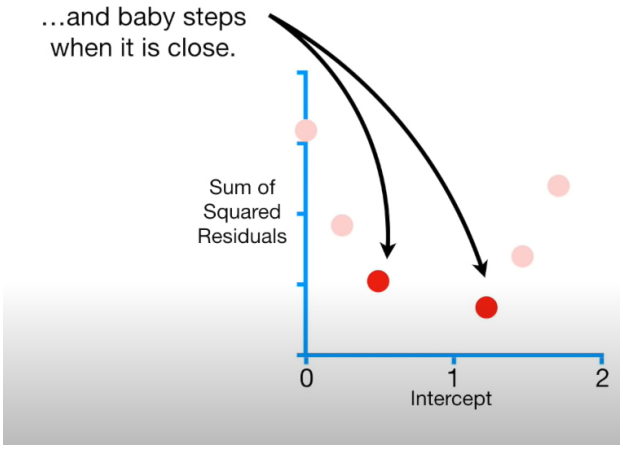
Slides 65 to 69: Big steps when far away, baby steps when close



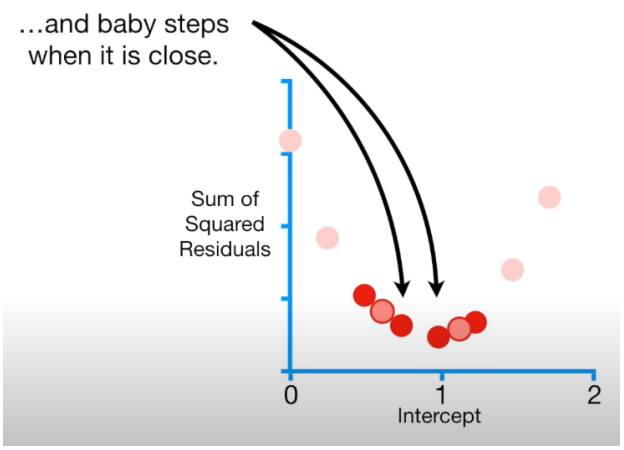
Slide 65



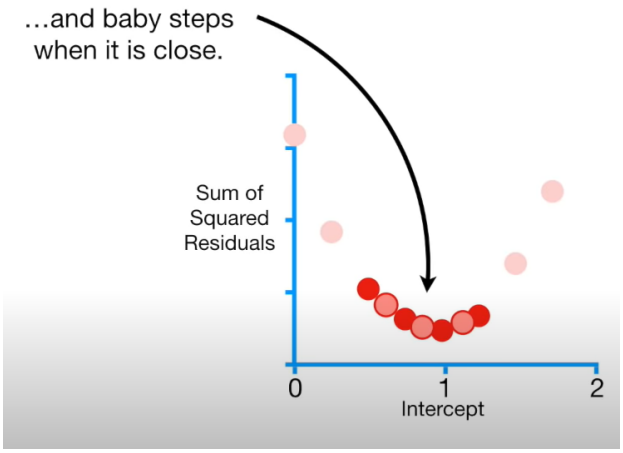
Slide 66



Slide 67



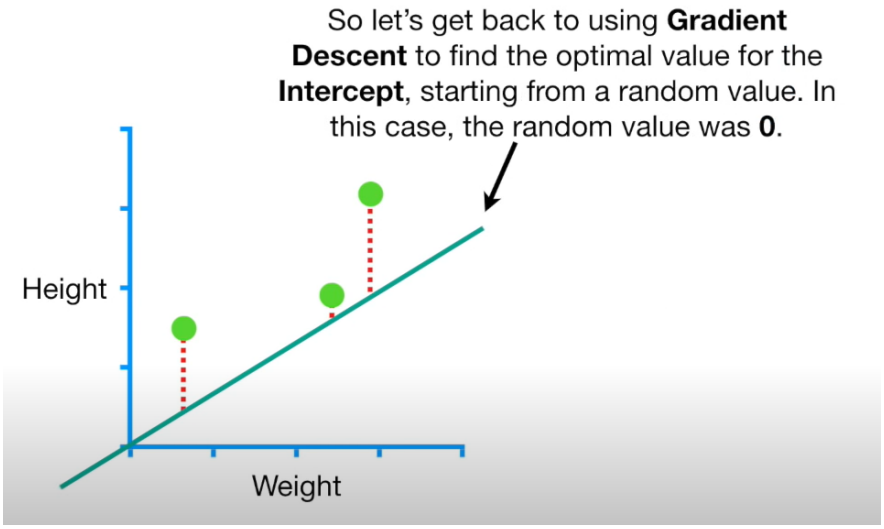
Slide 68



Slide 69

In one sentence: gradient descent finds the best intercept by **taking big steps when it is far from the answer and baby steps when it is close**. The dots on the SSR curve show this: they are spread out far from the bottom and crowd together near it.

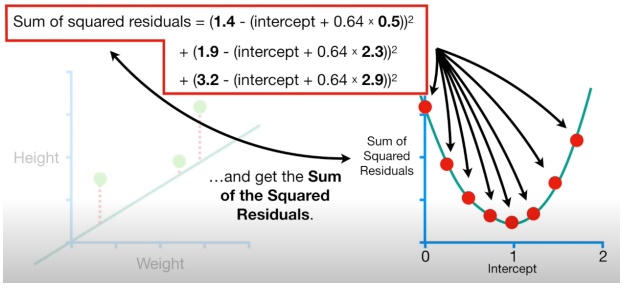
Slide 70: Back to our example



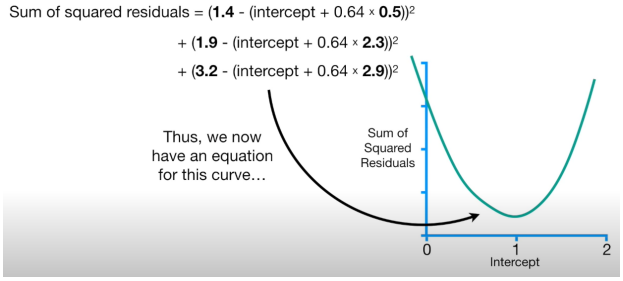
Slide 70

Back to the real problem: use gradient descent to find the best intercept, starting from a random value. In our case the random starting value was **0**.

Slides 71 to 72: Turning the dots into an equation



Slide 71

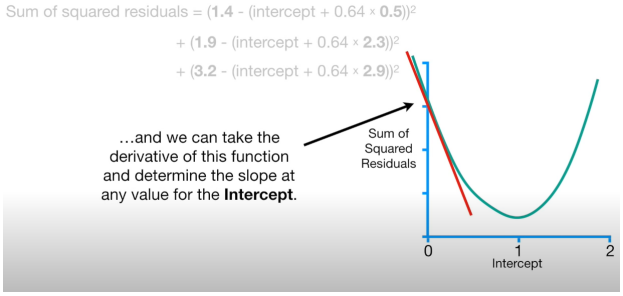


Slide 72

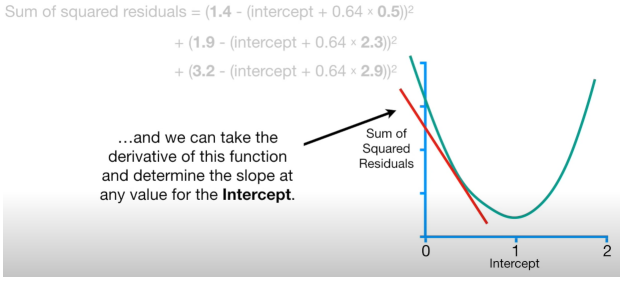
Each point on the SSR curve came from the same formula: **add up the squared residuals** for the three data points. Written out with the intercept left as an unknown, the SSR is:

- $(1.4 - (\text{intercept} + 0.64 \times 0.5))^2$
- $+ (1.9 - (\text{intercept} + 0.64 \times 2.3))^2$
- $+ (3.2 - (\text{intercept} + 0.64 \times 2.9))^2$
- Each bracket is one data point: observed height minus predicted height, squared

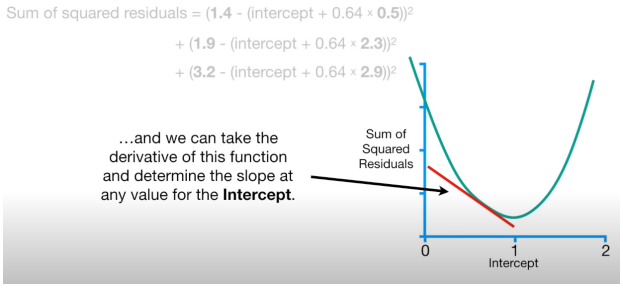
Slides 73 to 77: Taking the slope of the curve



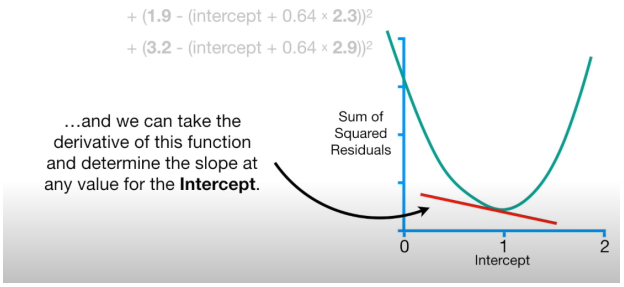
Slide 73



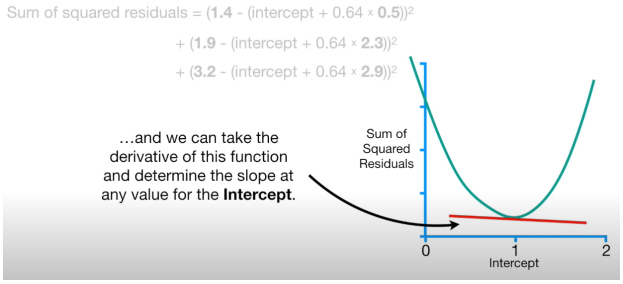
Slide 74



Slide 75



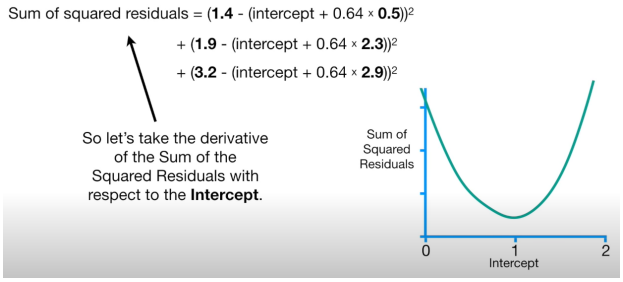
Slide 76



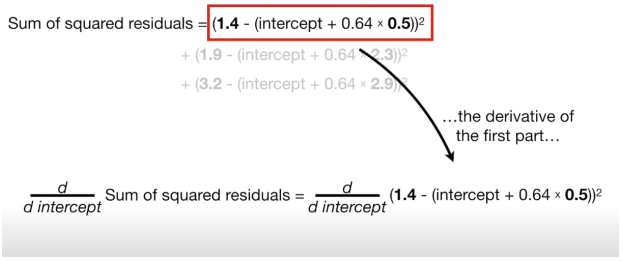
Slide 77

Because the SSR is now an equation, we can take its **derivative**. The derivative tells us the **slope of the curve** at any intercept. The red line touching the curve is the slope at that point. At the left the curve is steep (big negative slope); near the bottom the slope flattens to almost zero. **That slope is what tells gradient descent which way to step and how big the step should be.**

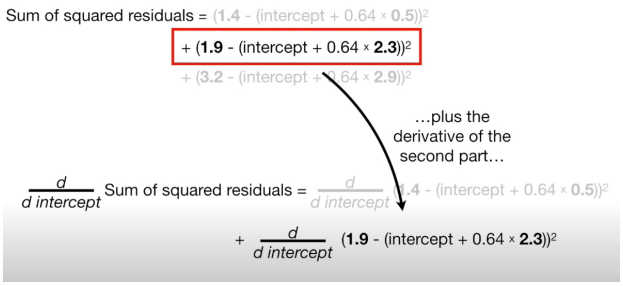
Slides 78 to 82: Take the derivative part by part



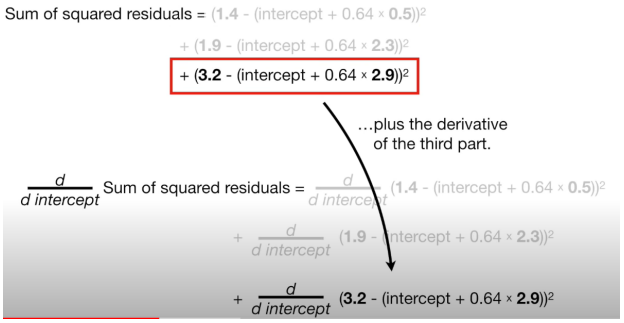
Slide 78



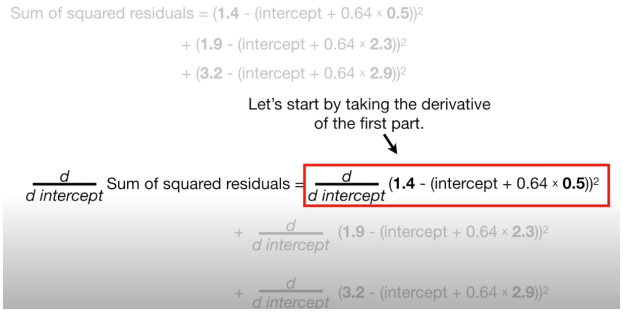
Slide 79



Slide 80



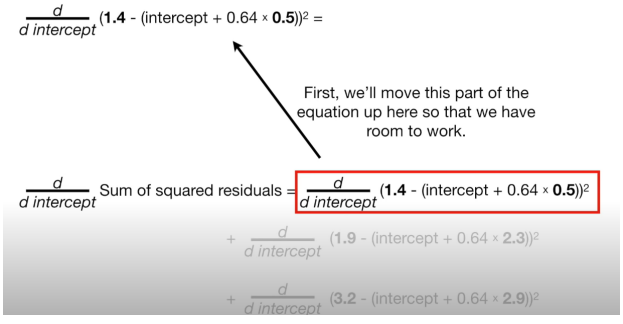
Slide 81



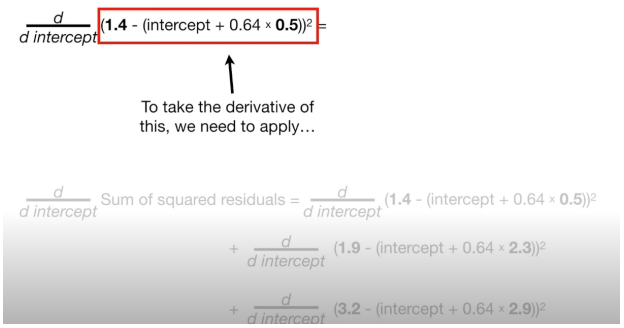
Slide 82

The SSR has three parts (one per data point). To differentiate it, differentiate each part separately and add the results: first the derivative of the first part (slide 79), then the second (slide 80), then the third (slide 81). The symbol **d/d intercept** means 'how fast does this change when the intercept changes'. Slide 82 starts with the first part.

Slides 83 to 84: Making room and choosing the tool



Slide 83



Slide 84

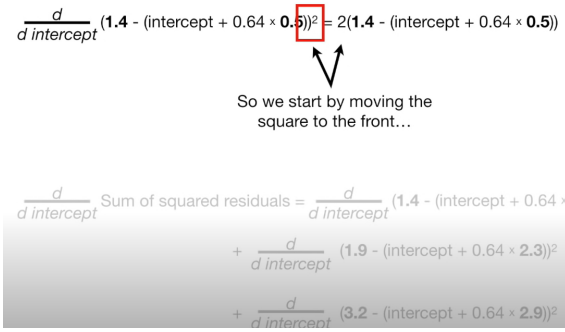
The first part is moved up so there is room to work. To differentiate something that has a bracket raised to a power, we need a rule called the **chain rule**.

Slide 85: The Chain Rule

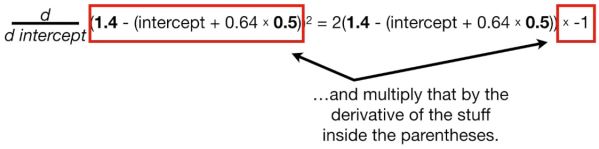
Slide 85

Title card for the next section. The **chain rule** is a maths rule for finding the derivative of a function that sits inside another function (here: a bracket inside a square). Steps are shown on the next slides.

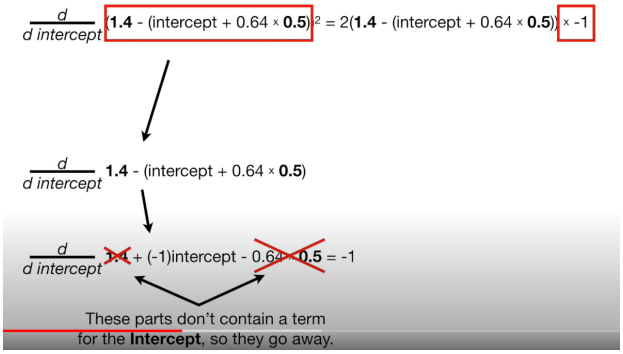
Slides 86 to 88: Applying the chain rule



Slide 86



Slide 87

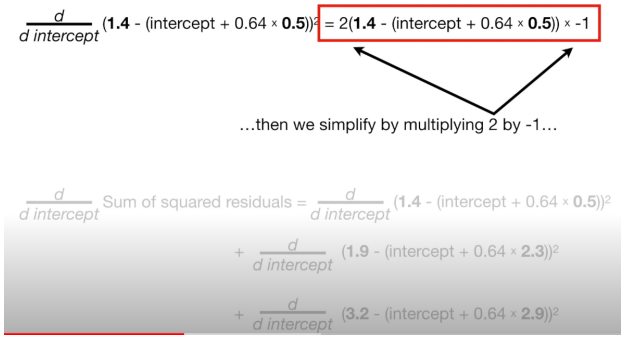


Slide 88

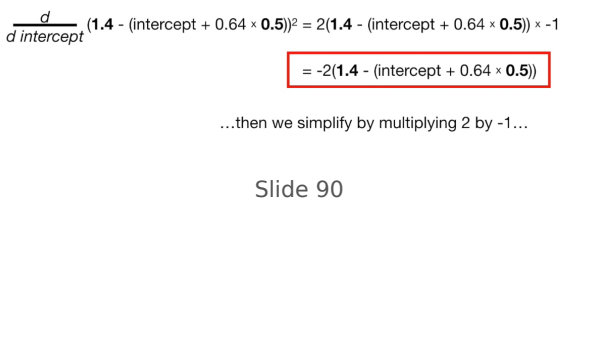
Two steps. **Step 1:** bring the square down to the front as a 2 (so (stuff)² becomes 2 × (stuff)). **Step 2:** multiply by the derivative of the stuff inside the brackets. Inside the bracket, 1.4 − (intercept + 0.64 × 0.5), only the intercept changes with the intercept, so the derivative of the inside is **−1** (the numbers 1.4 and 0.64 × 0.5 do not contain the intercept, so they disappear).

$$\frac{d}{d\,\text{intercept}}(1.4 - (\text{intercept} + 0.64 \times 0.5))^2 = 2(1.4 - (\text{intercept} + 0.64 \times 0.5)) \times (-1)$$

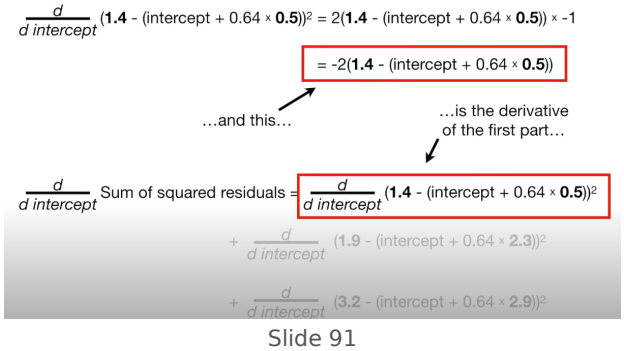
Slides 89 to 92: Simplify: the derivative of the first part



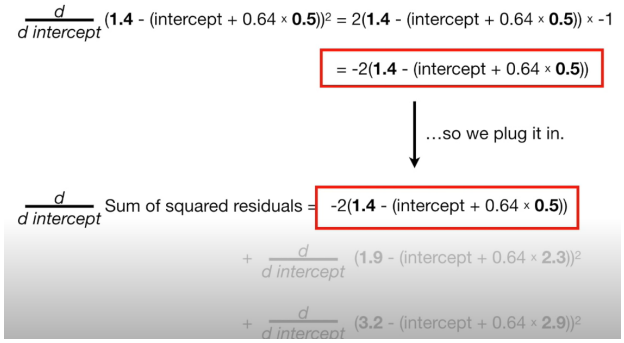
Slide 89



Slide 90



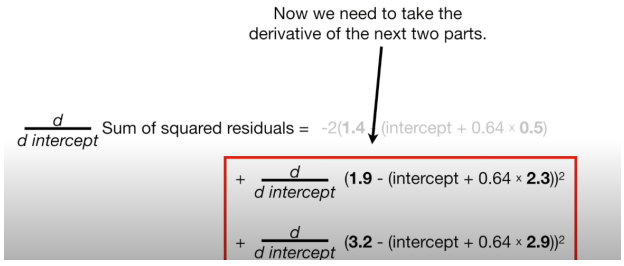
Slide 91



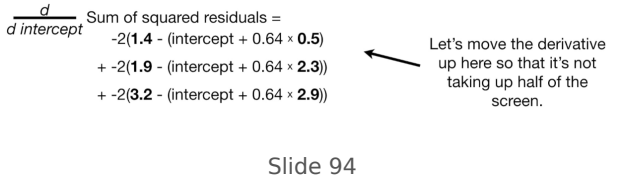
Slide 92

Multiply the 2 by the −1 and the first part's derivative is **−2 × (1.4 − (intercept + 0.64 × 0.5))**. That expression is now plugged back into the full derivative in place of the first part.

Slides 93 to 94: Same trick for the other two parts



Slide 93

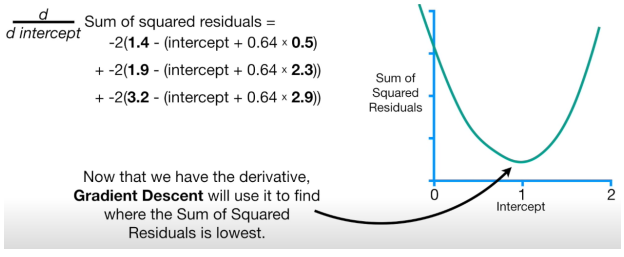


Slide 94

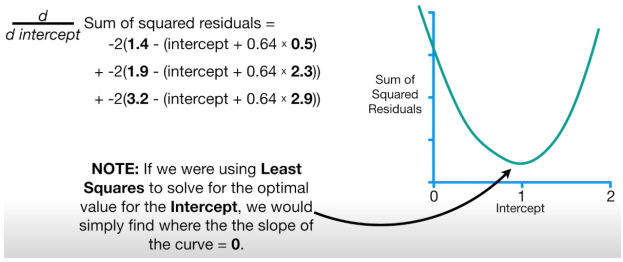
The second and third parts are handled in exactly the same way, so the video skips the working. After simplifying, the derivative of the whole Sum of Squared Residuals is the sum of three terms:

- −2 × (1.4 − (intercept + 0.64 × 0.5))
- −2 × (1.9 − (intercept + 0.64 × 2.3))
- −2 × (3.2 − (intercept + 0.64 × 2.9))

Slides 95 to 96: What the derivative is for



Slide 95

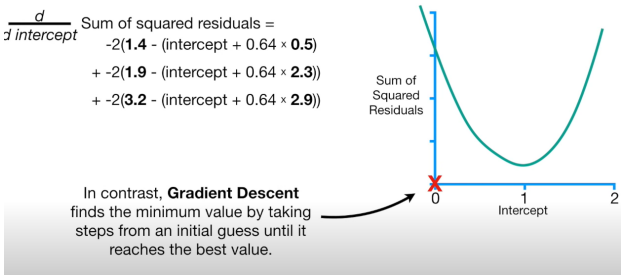


Slide 96

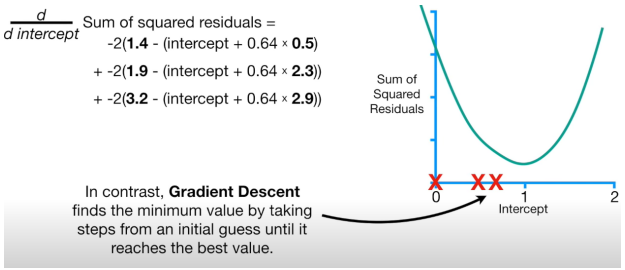
Now that we have the derivative, gradient descent will use it to find where the Sum of Squared Residuals is **lowest**. A side note: if we were using **Least Squares** instead of gradient descent,

we would just set the derivative equal to 0 and solve, because the slope of the curve is 0 at the very bottom.

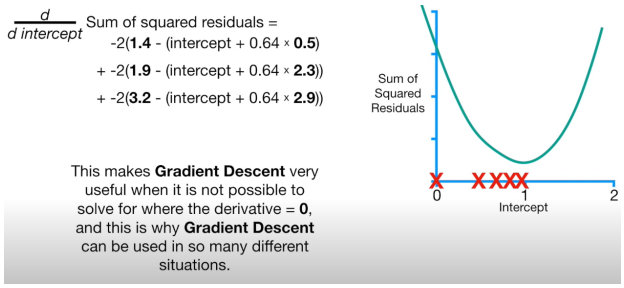
Slides 97 to 100: Why gradient descent, if Least Squares can solve it directly?



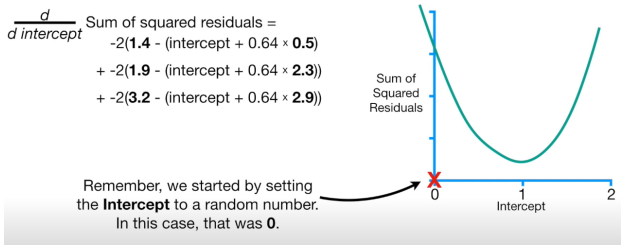
Slide 97



Slide 98



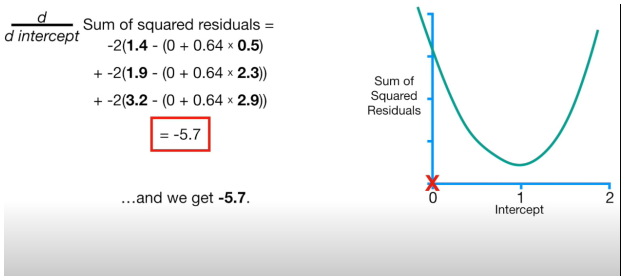
Slide 99



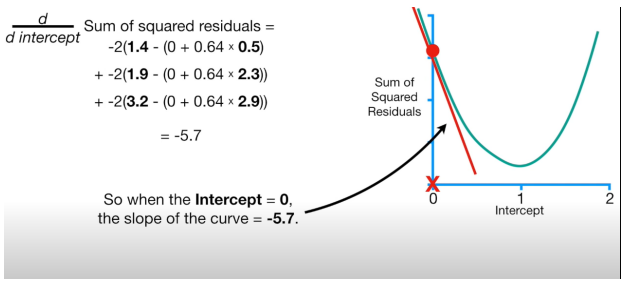
Slide 100

Least Squares only works for problems where you can solve 'derivative = 0' by hand. In many problems that is **not possible**. Gradient descent does not need that: it starts from an initial guess and takes steps until it reaches the best value. That is why it can be used in so many different situations. Remember: our starting guess for the intercept was **0**.

Slides 101 to 102: Slope at the starting guess



Slide 101

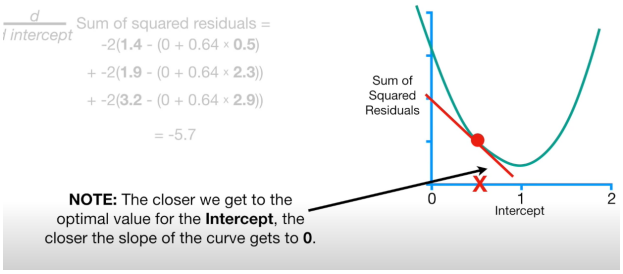


Slide 102

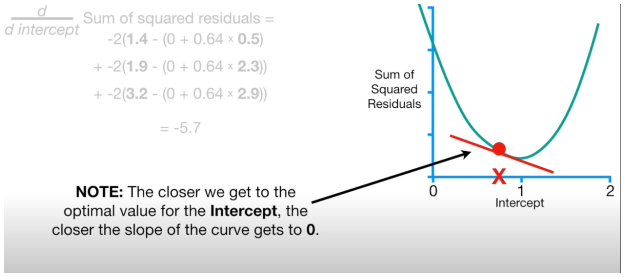
Plug the intercept **0** into the derivative. The three terms add up to **−5.7**. So when the intercept is 0, the slope of the SSR curve is **−5.7**. The negative sign means the curve is going downhill as the intercept grows. The red line on the curve shows that steep slope.

$$\text{slope of SSR at intercept } 0 = -5.7$$

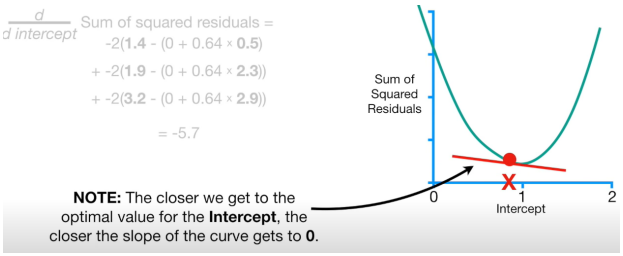
Slides 103 to 107: Near the bottom, the slope is close to zero



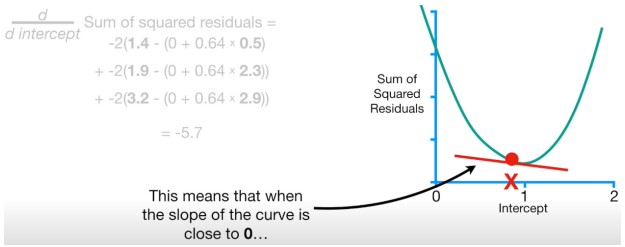
Slide 103



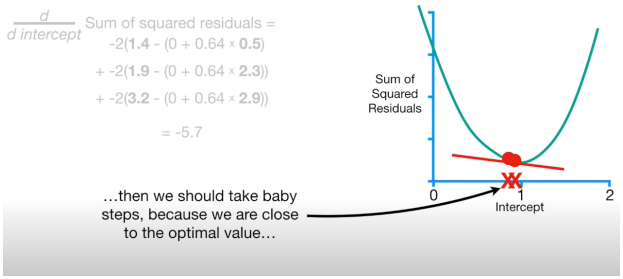
Slide 104



Slide 105



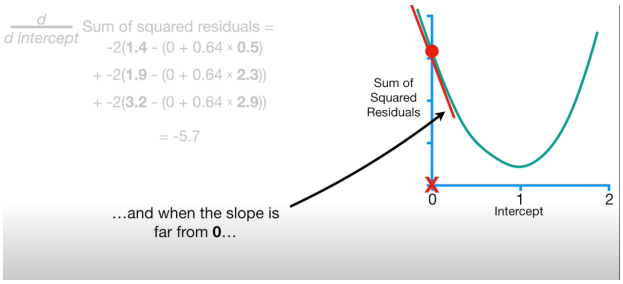
Slide 106



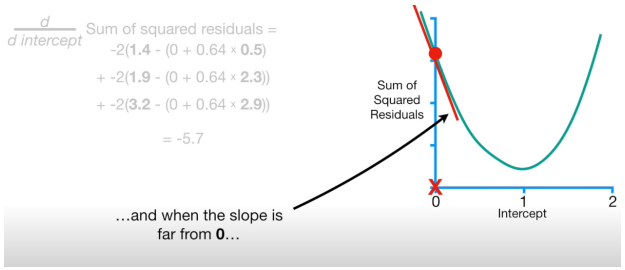
Slide 107

Look at the red tangent line as the guess moves towards the lowest point: it gets flatter and flatter. **The closer we get to the optimal intercept, the closer the slope gets to 0.** So a slope close to 0 tells us we are close to the answer and should take **baby steps**.

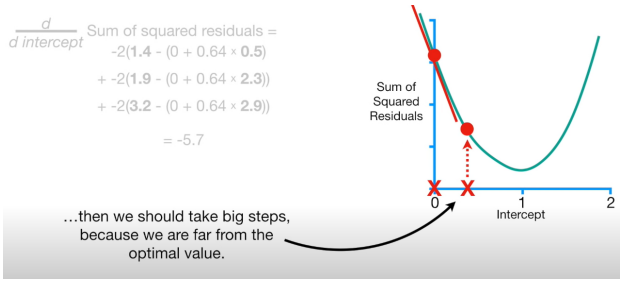
Slides 108 to 111: Far from the bottom, the slope is steep



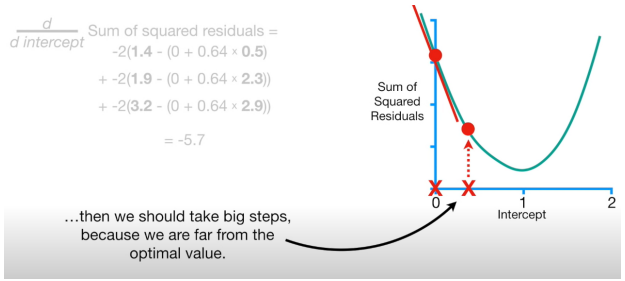
Slide 108



Slide 109



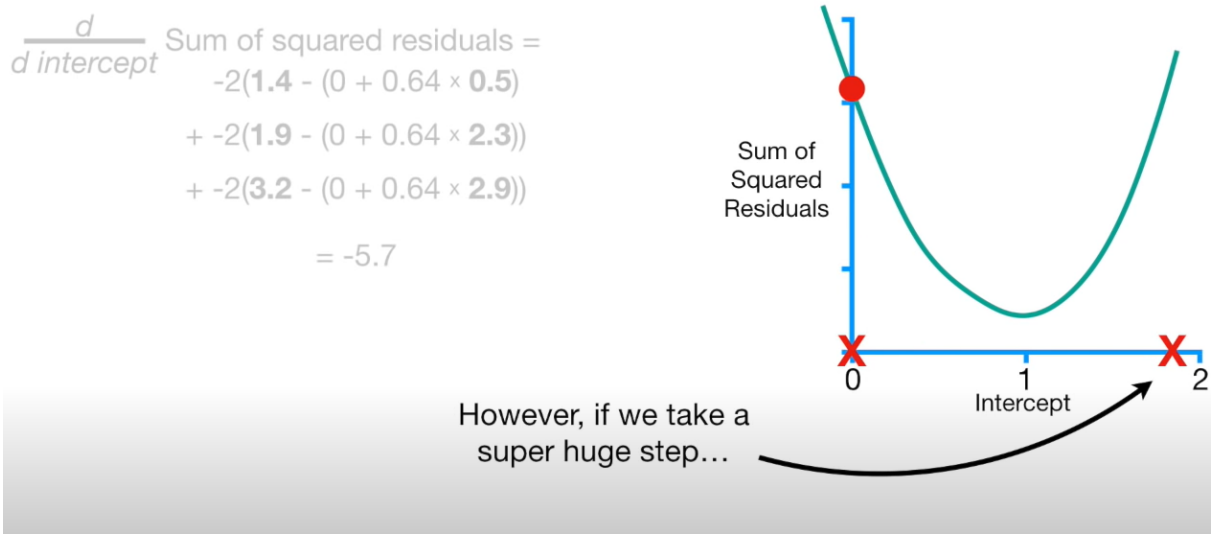
Slide 110



Slide 111

When the slope is far from 0 (a steep red line), we are far from the optimal value, so we should take **big steps**. In short: **the size of the step is tied to the size of the slope.**

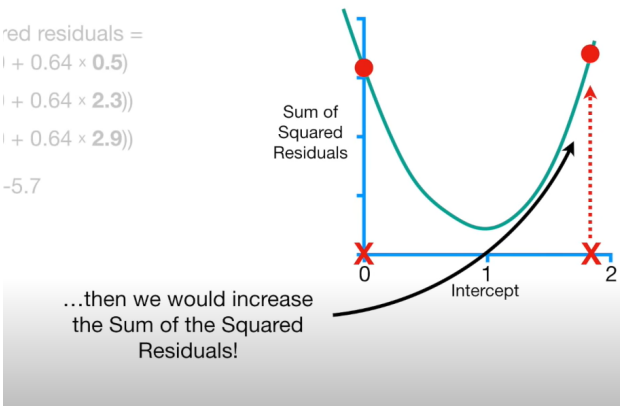
Slide 112: But a huge step can overshoot



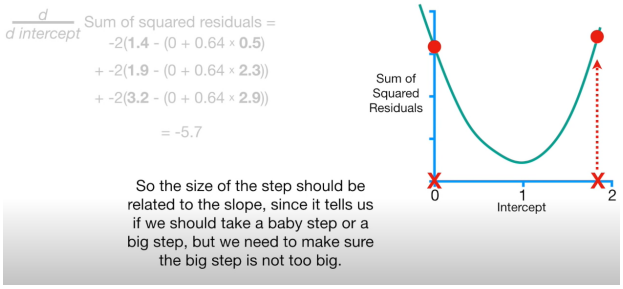
Slide 112

If we took a super huge step, we could jump right past the lowest point to the far side of the curve (the red dot at the right edge). So the step size has to be controlled, and that is where the **learning rate** comes in on the next slides.

Slides 113 to 114: The step size must be tied to the slope, but not too big



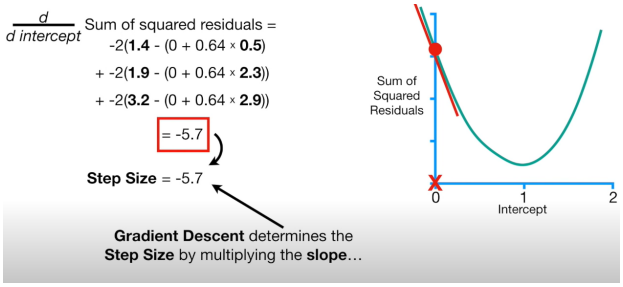
Slide 113



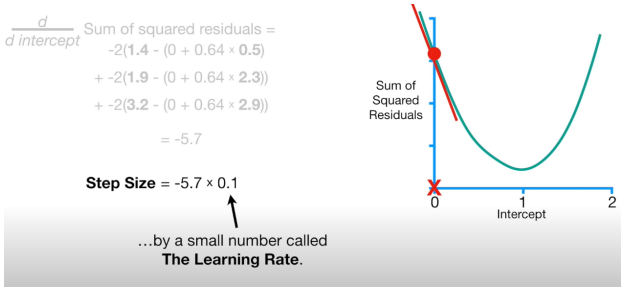
Slide 114

If the step is too huge, we jump past the bottom and the Sum of Squared Residuals actually **goes up** again. So the step size should be **related to the slope** (steep = big step, flat = baby step), while making sure the big step is **not too big**.

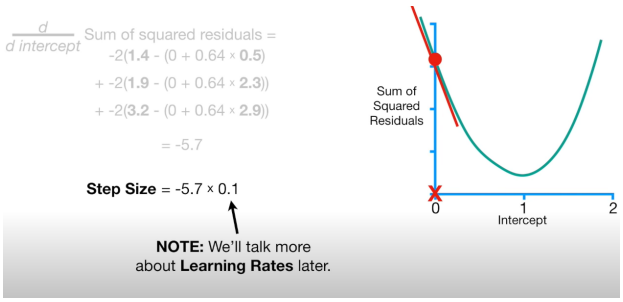
Slides 115 to 117: Step size = slope × learning rate



Slide 115



Slide 116

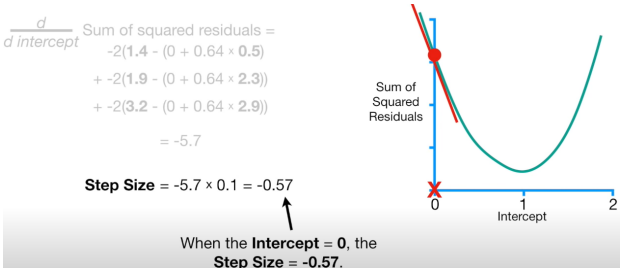


Slide 117

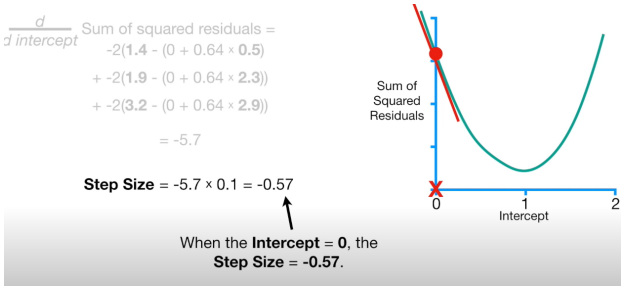
Gradient descent works out the **Step Size** by **multiplying the slope by a small number called the Learning Rate**. Here the slope is −5.7 and the learning rate is 0.1, so **Step Size = −5.7 × 0.1**. (Learning rates are discussed in more detail later.)

$$\text{Step size} = \text{slope} \times \text{learning rate}$$

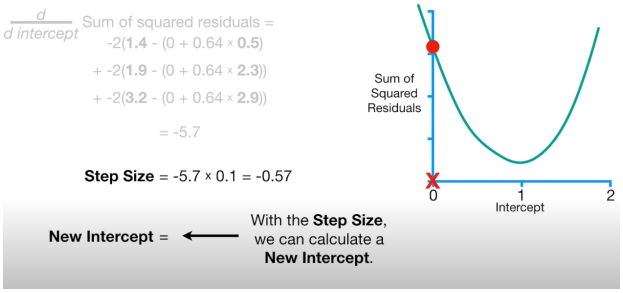
Slides 118 to 120: The first step size



Slide 118



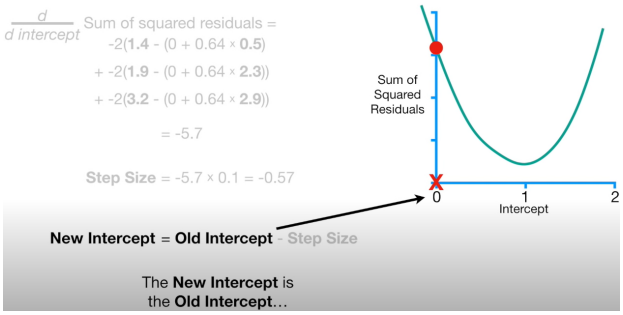
Slide 119



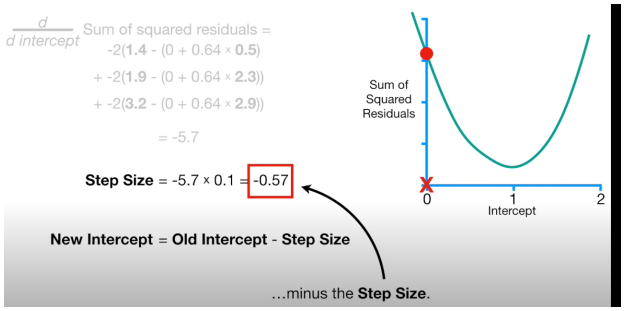
Slide 120

Step Size = −5.7 × 0.1 = **−0.57**. So when the intercept is 0, the step size is −0.57.

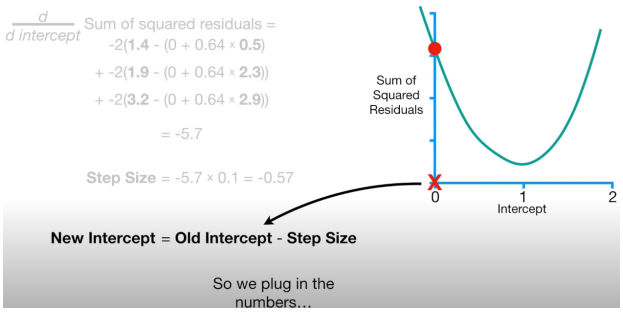
Slides 121 to 127: Take the step: the new intercept



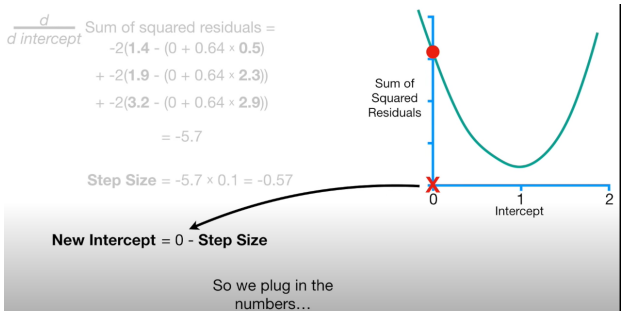
Slide 121



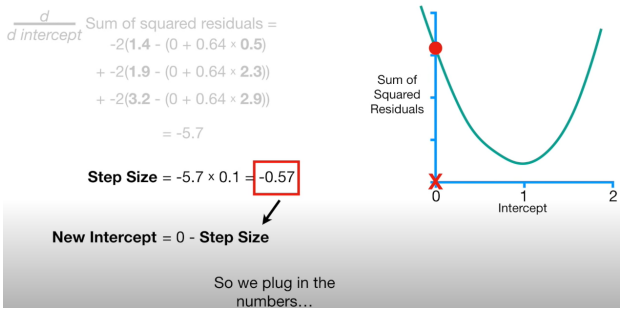
Slide 122



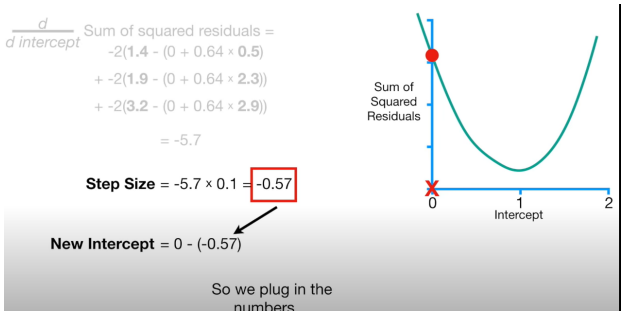
Slide 123



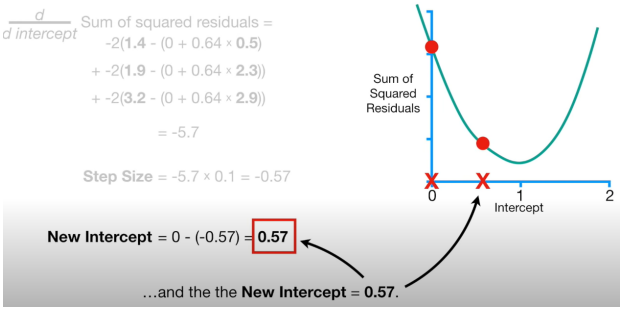
Slide 124



Slide 125



Slide 126

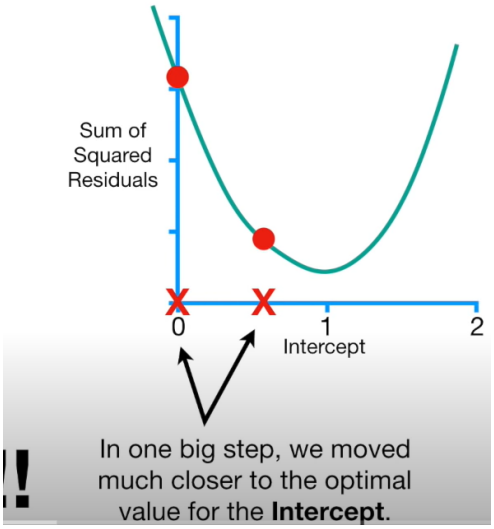


Slide 127

With the step size we calculate the **new intercept**: **New Intercept = Old Intercept − Step Size**. Plugging in the numbers: $0 - (-0.57) = \mathbf{0.57}$. The red dot moves from intercept 0 to intercept 0.57, a good bit closer to the bottom of the curve. (Subtracting a negative step moves the intercept up; this is why the sign works out.)

$$\text{New intercept} = \text{Old intercept} - \text{Step size} = 0 - (-0.57) = 0.57$$

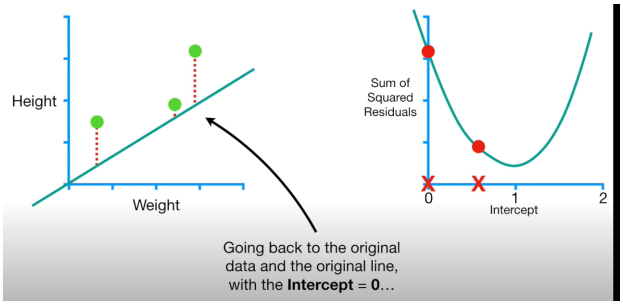
Slide 128: One big step took us far



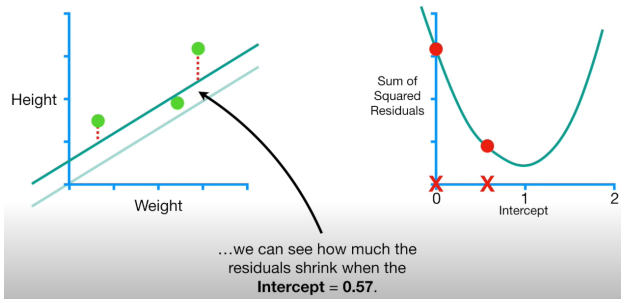
Slide 128

Comparing the first point (intercept 0) and the new point (intercept 0.57): in **one big step** we moved much closer to the optimal value of the intercept.

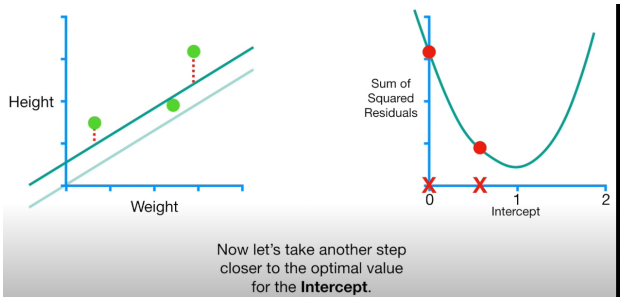
Slides 129 to 131: What the move did to the line



Slide 129



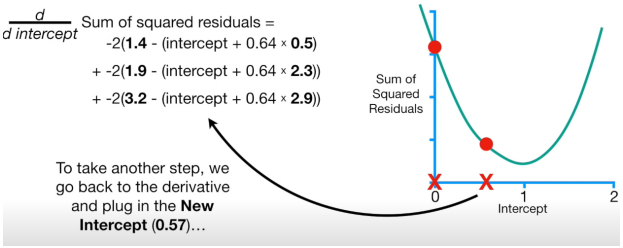
Slide 130



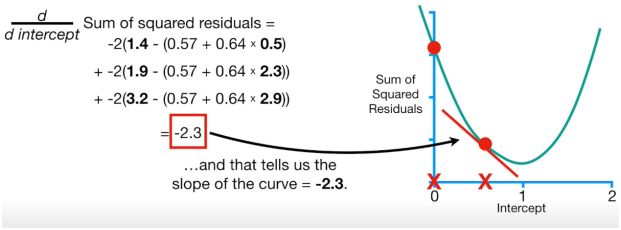
Slide 131

Going back to the data. The darker line is the original line (intercept 0); the lighter line shows the new line with intercept 0.57. The new line sits closer to the green points, so the residuals (dotted red lines) have shrunk.

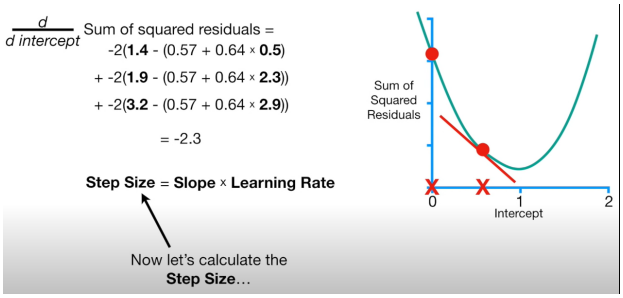
Slides 132 to 136: Take another step



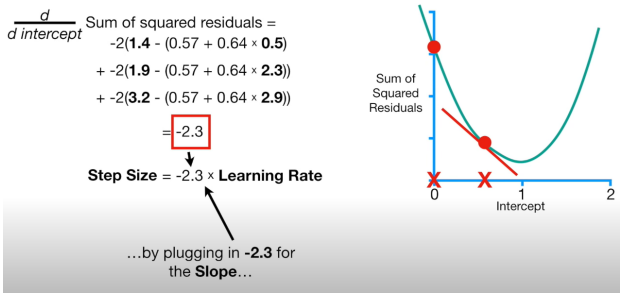
Slide 132



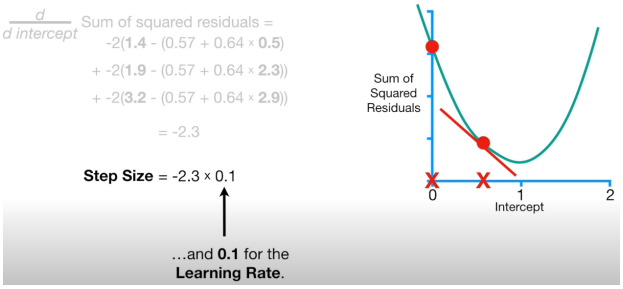
Slide 133



Slide 134



Slide 135

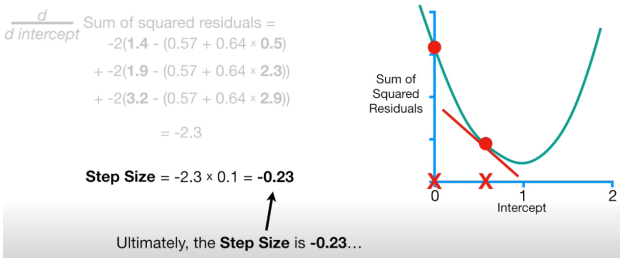


Slide 136

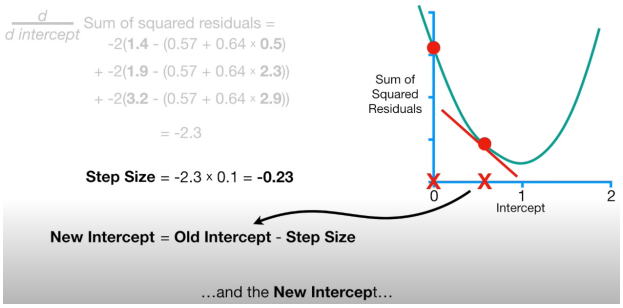
Now repeat the whole process from the new intercept, 0.57. Plug 0.57 into the derivative: the slope of the curve there is **−2.3**. That is flatter than before (−5.7), because we are closer to the bottom. The step size is slope × learning rate = **−2.3 × 0.1**.

- Same recipe each time: slope → step size → new intercept
- The slope is smaller, so the step will be smaller: 'baby steps when close'

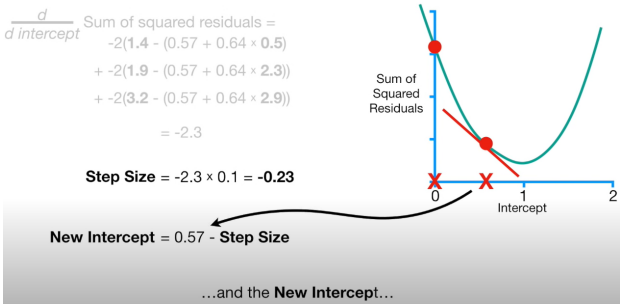
Slides 137 to 141: Second step: 0.57 becomes 0.8



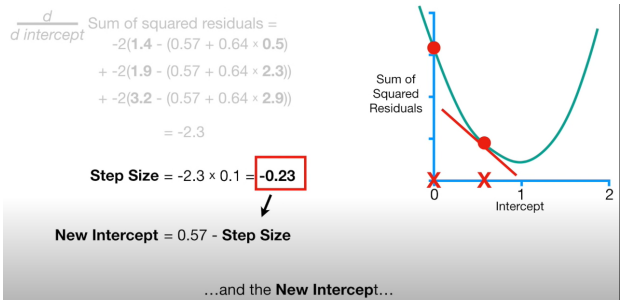
Slide 137



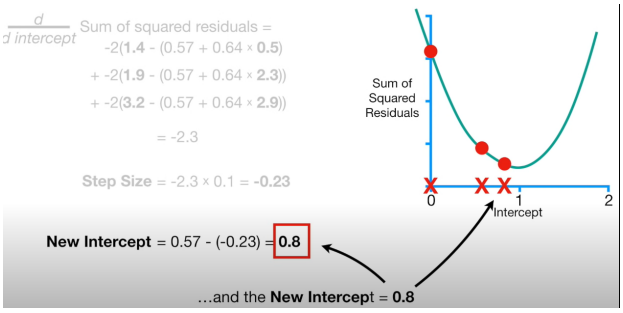
Slide 138



Slide 139



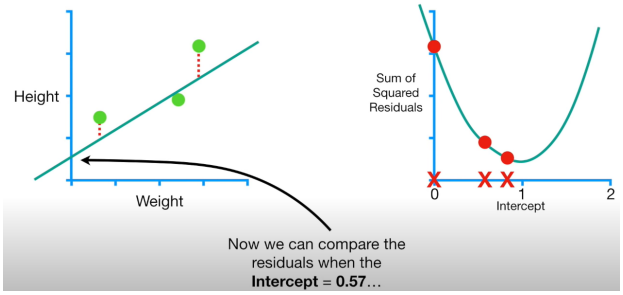
Slide 140



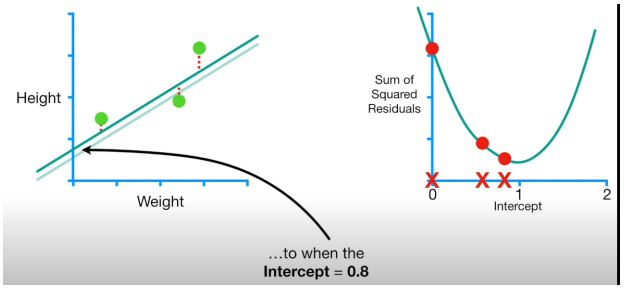
Slide 141

Step size = $-2.3 \times 0.1 = \mathbf{-0.23}$. New intercept = $0.57 - (-0.23) = \mathbf{0.8}$. The dot moves a bit further down the curve.

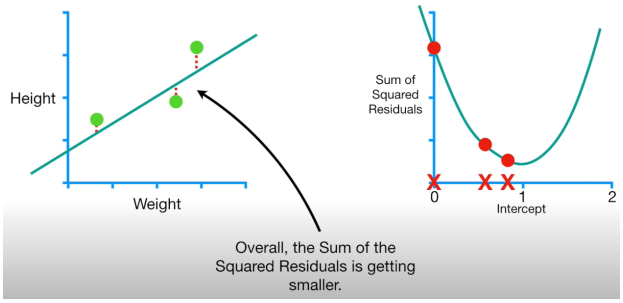
Slides 142 to 145: Compare: the second step was smaller than the first



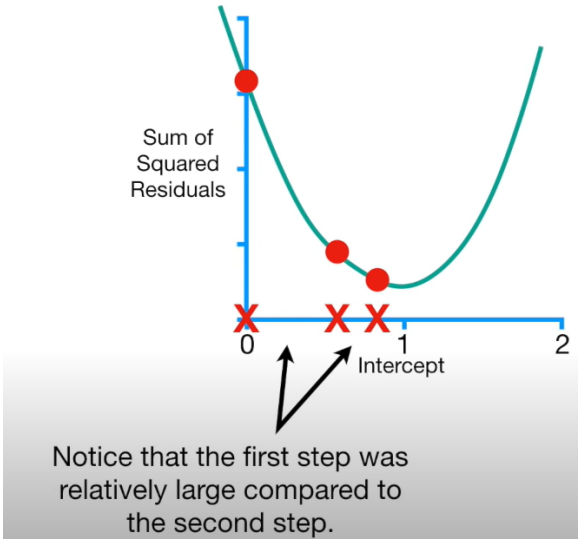
Slide 142



Slide 143



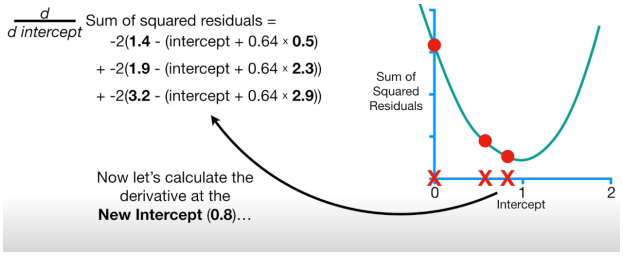
Slide 144



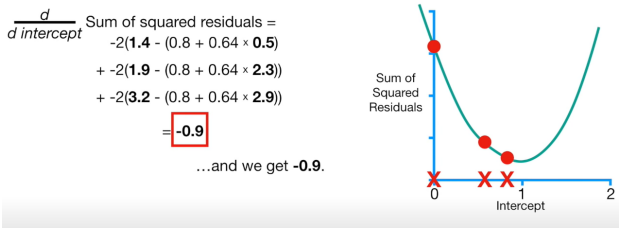
Slide 145

Overall the Sum of Squared Residuals is getting smaller: the line is closer to the points. Also notice that **the first step (0 → 0.57) was big compared with the second step (0.57 → 0.8)**, because the slope was steeper at the start.

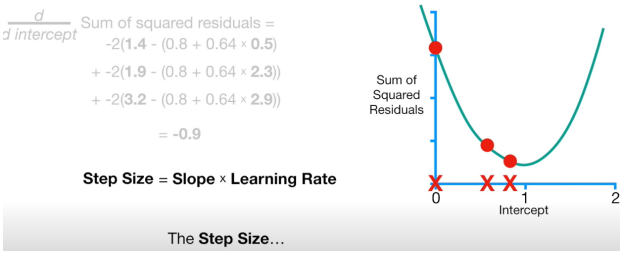
Slides 146 to 151: Third step: 0.8 becomes 0.89



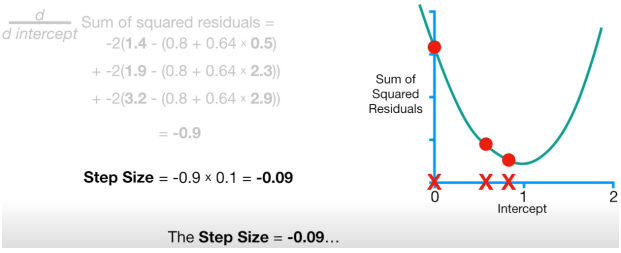
Slide 146



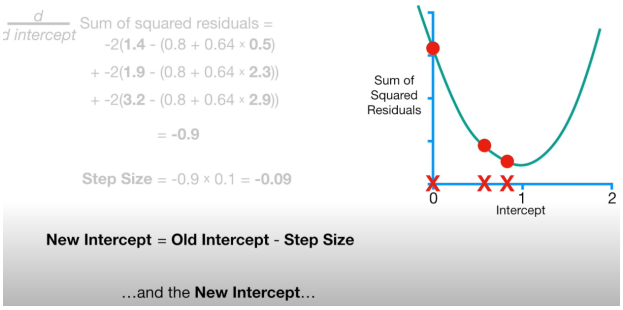
Slide 147



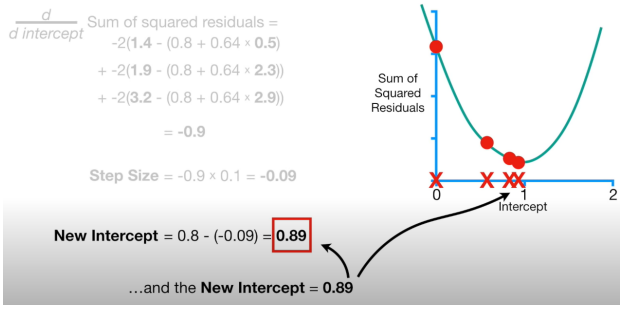
Slide 148



Slide 149



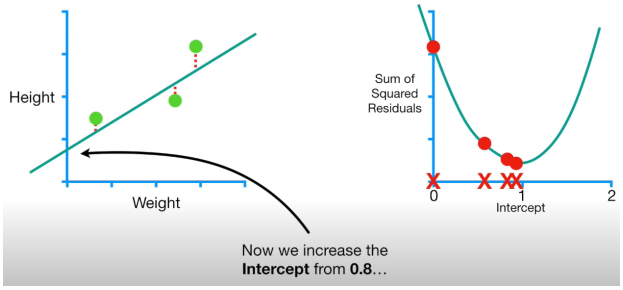
Slide 150



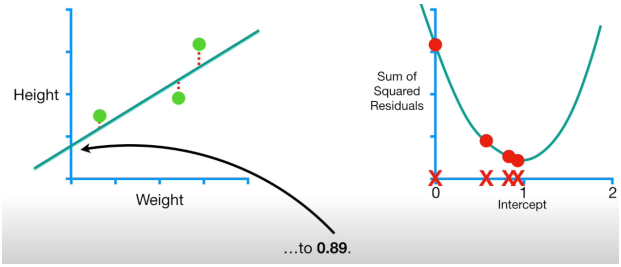
Slide 151

Repeat once more. Plug 0.8 into the derivative: slope **−0.9**. Step size = −0.9 × 0.1 = **−0.09**. New intercept = 0.8 − (−0.09) = **0.89**. The step has become very small.

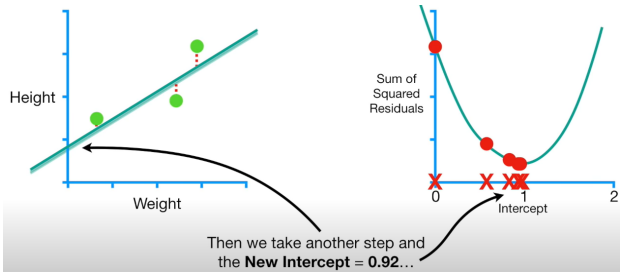
Slides 152 to 156: More steps: 0.89 → 0.92 → 0.94 → 0.95



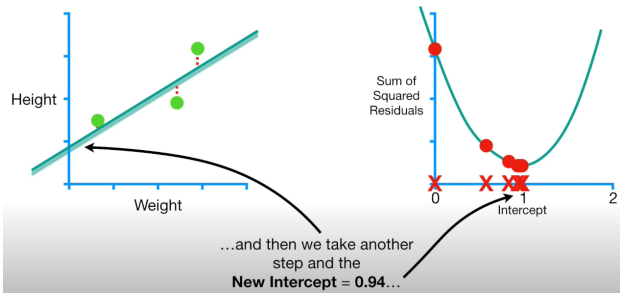
Slide 152



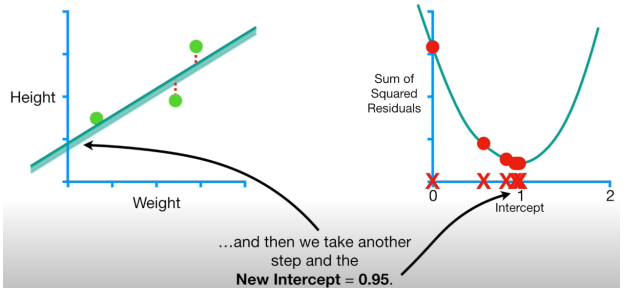
Slide 153



Slide 154



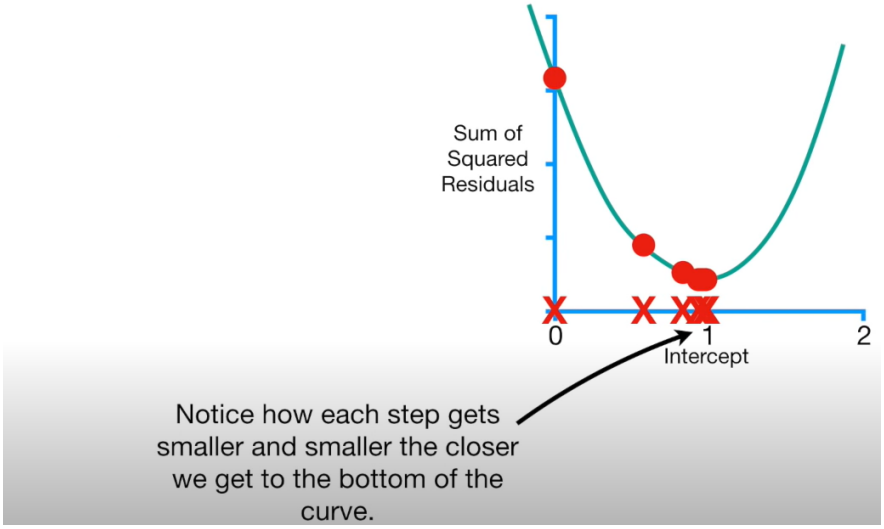
Slide 155



Slide 156

The same recipe keeps being applied. The line is shifted up by smaller and smaller amounts: the intercept goes from 0.89 to 0.92, then 0.94, then 0.95. The lighter and darker lines on the left almost overlap, meaning we have almost stopped improving.

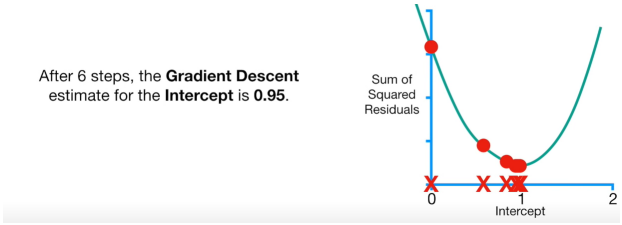
Slide 157: Steps shrink as we near the bottom



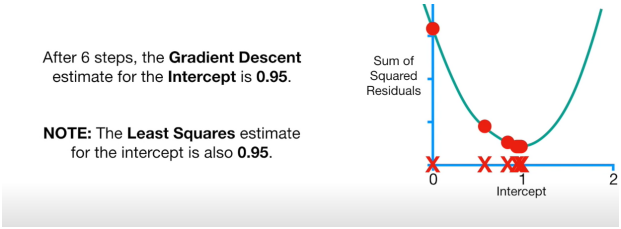
Slide 157

Notice how **each step gets smaller and smaller the closer we get to the bottom of the curve** (the red crosses on the intercept axis bunch together near 1).

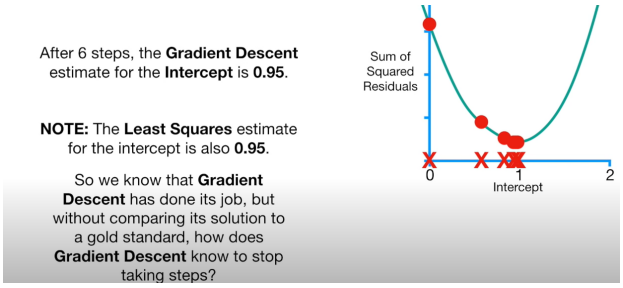
Slides 158 to 160: The answer after 6 steps: 0.95



Slide 158



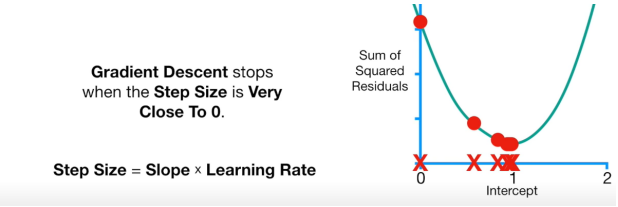
Slide 159



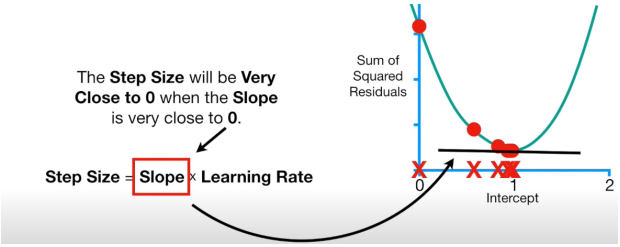
Slide 160

After 6 steps, gradient descent estimates the intercept to be **0.95**. As a check, the **Least Squares** method (which solves the problem exactly) also gives **0.95**. So gradient descent found the right answer. But in real problems there is no gold standard to compare with, so how does gradient descent know when to stop?

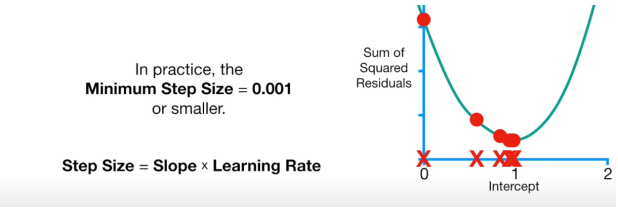
Slides 161 to 167: Stopping rule 1: the step size becomes tiny



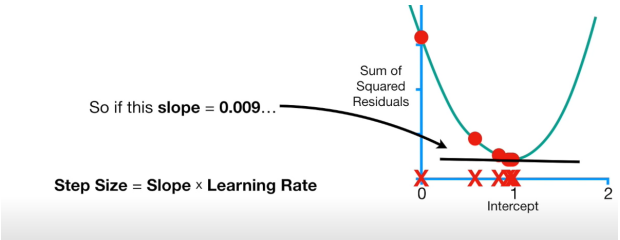
Slide 161



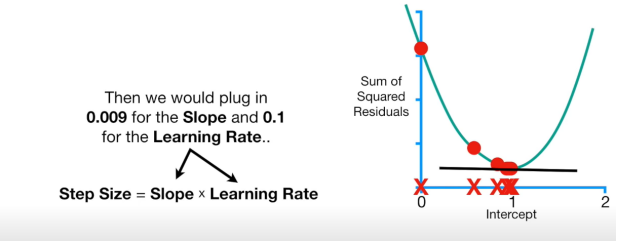
Slide 162



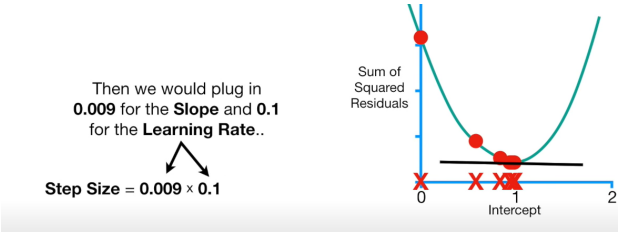
Slide 163



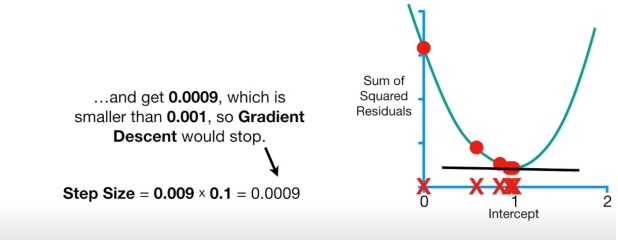
Slide 164



Slide 165



Slide 166

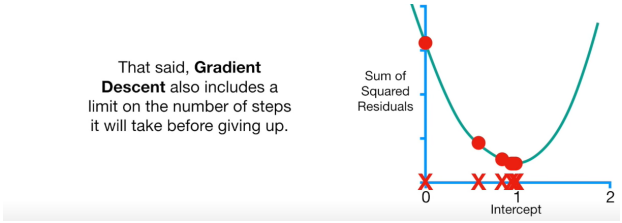


Slide 167

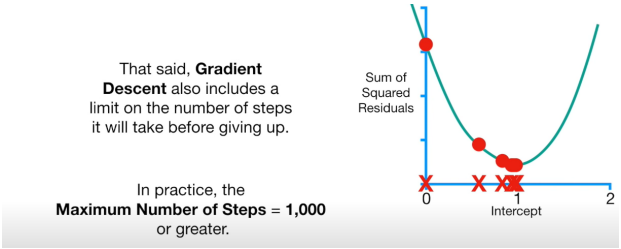
Gradient descent stops when the step size is very close to 0. Since step size = slope × learning rate, a step size near 0 means the slope is near 0, which means we are at the bottom of the curve. In practice, the **minimum step size is 0.001 or smaller**. In the example, if the slope is 0.009 and the learning rate 0.1, the step size is 0.009 × 0.1 = 0.0009. That is smaller than 0.001, so gradient descent stops.

$$0.009 \times 0.1 = 0.0009 < 0.001 \Rightarrow \text{stop}$$

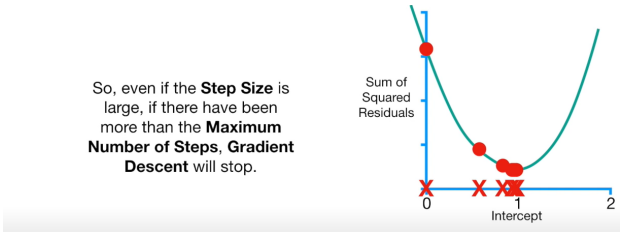
Slides 168 to 170: Stopping rule 2: a limit on the number of steps



Slide 168



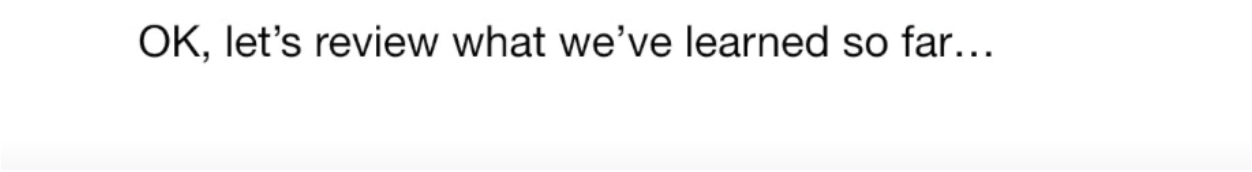
Slide 169



Slide 170

Gradient descent also has a **maximum number of steps**. In practice this is **1,000 or greater**. Even if the step size is still large, once the maximum is reached gradient descent stops anyway. This stops a badly behaved search from running forever.

Slide 171: End of this part: a review



Slide 171

The video ends this section with 'OK, let's review what we've learned so far…'. The recipe you just saw, in one list, is below.

- 1. Choose a loss function (here: Sum of Squared Residuals)
- 2. Pick a random starting value for the number you are optimising (here the intercept, 0)
- 3. Work out the slope of the loss at that value (the derivative)
- 4. Step size = slope × learning rate
- 5. New value = old value − step size
- 6. Repeat until the step size is tiny, or the maximum number of steps is reached